

Capstone Project



Final Documentation

End-to-End Salesforce CRM Transformation

for Customer Operations and Service Management

Author: Syakila Aini

M1. Project Initiation and Stakeholder Analysis

1.1 Project Charter

1.1.1 Project Overview

GreenGrid Utilities is a mid-sized provider of energy and utility services facing substantial operational challenges as a result of a highly disjointed system environment. The organization currently depends on numerous disconnected legacy platforms, manual data entry practices, and siloed spreadsheets to administer core customer records, coordinate field dispatch activities, and manage on-site operations.

This fragmented infrastructure leads to redundant data entries, prolonged case resolution timelines, and limited real-time visibility for senior management into operational performance. To comprehensively address these challenges, GreenGrid Utilities is undertaking a strategic transition to Salesforce as a unified Customer Relationship Management (CRM) platform. This initiative brings together customer operations, internal service desk functions, and infrastructure dispatch coordination under a single, authoritative system.

1.1.2 Problem Statement

The existing technology environment at GreenGrid Utilities lacks the capability to support cohesive, cross-departmental collaboration. Customer records are scattered and duplicated across regional repositories, field dispatch processes depend entirely on verbal communication, and executive leadership has no immediate access to dependable service delivery performance data.

The absence of a centralized system makes it increasingly difficult for the organization to consistently deliver high-quality, responsive customer service.

1.1.3 Project Scope

The project boundaries are clearly defined to maintain control over delivery and prevent unplanned scope expansion. To meet the necessary data management requirements without reliance on live external system feeds, legacy operational data will be imported directly into Salesforce using standard data loading utilities.

In Scope	Out of Scope
Consolidated customer data governance and account profiles	Financial administration and utility billing systems
Sales pipeline management and account relationship tracking	Real-time synchronization with external third-party platforms
Customer service support and phone-based case documentation workflows	Customer-facing self-service digital portals
Field service scheduling, dispatch coordination, and technician task tracking	Proprietary mobile application development

1.2 Stakeholder Identification

Stakeholders represent the individuals and organizational groups who are directly engaged in, or meaningfully affected by, the Salesforce CRM implementation initiative.

1.2.1 Internal Stakeholders

- Executive Management: Senior leadership requiring clear and consolidated operational data views.
- Customer Support Team: Front-line service agents responsible for processing incoming customer requests and issue reports.
- Field Technicians: Specialized engineers who carry out physical utility maintenance and on-site service tasks.
- Sales Team: Business development representatives managing client account acquisition and relationship activities.
- IT / System Administrator: Technical personnel overseeing platform configuration, governance, and security.

1.2.2 External Stakeholders

- Customers: End-users who receive local utility distribution services and are directly impacted by service quality.

Each identified stakeholder group carries distinct operational priorities, day-to-day responsibilities, and degrees of influence over the platform configuration and implementation approach.

1.3 Stakeholder Analysis Matrix

The matrix below assesses GreenGrid Utilities stakeholders based on their degree of organizational authority (Power) and their level of operational impact and engagement (Interest) relative to the CRM transformation initiative.

Stakeholder Group	Role in Organization	Interest Level	Power Level	Key Needs and Operational Concerns
Executive Management	Strategic decision-makers	High	High	Live analytical dashboards and integrated operational performance reporting.
Customer Support Team	Front-line service desk representatives	High	Medium	Efficient data retrieval tools and a consolidated, intuitive case management interface.
Field Technicians	On-site infrastructure service engineers	High	Medium	Organized task records, accurate location data, and elimination of paper-based administrative processes.
Sales Team	Client acquisition and account managers	High	Medium	Unified client history records and thorough removal of duplicate account entries.
IT / System Admin	Platform governance and configuration specialists	Medium	High	Flexible data architecture, rule-based automation frameworks, and comprehensive access control.

Customers	End-users of utility distribution services	High	Low	Reduced wait times on support lines and prompt resolution of service interruptions.
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1.4 Business Objectives Definition

1.4.1 Defined Business Objectives

a. Streamline Customer Support Operations:

Reduce call center processing burden by accelerating data retrieval and simplifying case tracking workflows.

b. Ensure Data Accuracy and Consistency:

Implement a consolidated native data schema to systematically eliminate duplicate customer records and ensure data integrity across the organization.

c. Enable Real-Time Operational Visibility:

Equip executive leadership with up-to-date operational metrics through built-in reporting and dashboard capabilities.

d. Integrate Internal Communication Channels:

Establish seamless connectivity between customer service case logs and active field maintenance dispatch activities within a unified platform.

1.4.2 Success Indicators

a. Average Handling Time (AHT) Reduction:

A quantifiable decrease in the time required for service agents to record and process incoming customer issues.

b. Duplicate Record Elimination Target:

A measurable reduction in the number of redundant customer account and contact entries within the system.

c. Dashboard Utilization Rate:

Regular and sustained usage of live reporting tools by management teams as a primary decision-making resource.

M2. Current State Assessment and Requirements Gathering

2.1 Process Mapping

2.1.1 AS-IS Customer Service Process Flow

The existing AS-IS operational workflow for customer service at GreenGrid Utilities is characterized by heavy manual intervention, lack of optimization, and systemic disconnection between departments. A typical utility service incident follows the sequence outlined below:

- **Step 1 (Intake):** A customer encounters a localized utility emergency or power disruption and notifies the support office via a standard telephone call.
- **Step 2 (Lookup):** Upon receiving the call, the support agent performs a manual search for the customer's identity, account standing, and property location within an oversized Excel spreadsheet. This constitutes bottleneck B1.
- **Step 3 (Dispatch):** After retrieving the relevant data, the agent conveys case information to field personnel through informal, non-auditable channels such as direct calls or WhatsApp messaging. This constitutes bottleneck B2.
- **Step 4 (Execution):** Field technicians travel to the designated location, carry out physical infrastructure repairs, and record their service notes on paper-based forms.
- **Step 5 (Closeout):** Status updates are relayed back to the main office verbally or through text messages, after which the support agent manually inputs the information into the Excel file. This constitutes bottleneck B3.

2.1.2 Process Challenges (Bottleneck Analysis)

a. B1 (Manual Data Lookup):

Searching through legacy spreadsheet records demands approximately 3 to 5 minutes per incoming call, leading to elevated call abandonment rates and degraded customer satisfaction.

b. B2 (Manual Documentation & Communication Gaps):

The practice of coordinating field crew dispatch through informal text or voice-based applications, combined with paper-based technical logging, results in data fragmentation and a 15% human error rate during job detail transmission.

c. B3 (Data Latency):

Central spreadsheet records are typically updated 2 to 4 hours after physical job completion, which undermines management's ability to monitor grid status in real time.

2.2 Stakeholder Interviews

Simulated discovery sessions and user-centered interviews were conducted across multiple departments to collect qualitative operational insights:

- **Customer Support Team Input:** Support agents conveyed significant frustration with managing multiple disconnected tracking sheets, expressing a clear need for a consolidated console interface to eliminate constant file switching.
- **Field Operations Team Input:** Field crews indicated that the absence of direct access to customer interaction history or technical asset data results in avoidable delays, estimating that digital access could reduce return-trips by approximately 20%.
- **Executive Management Input:** Leadership emphasized the urgent need for automated, system-generated operational reporting to reclaim the 4 hours per week currently lost to manual data consolidation efforts.

2.3 System Requirements Documentation

2.3.1. Business Requirements (BR)

The following strategic mandates outline the overarching corporate objectives for the CRM implementation initiative.

ID	Business Requirement Description	Success Metric / Target
BR1	Reduce customer call waiting periods and improve the throughput of the call center handling process.	Attain an immediate 25% reduction in Average Handling Time (AHT).
BR2	Standardize the tracking of core infrastructure data to eliminate database fragmentation.	Establish a unified source of truth with a 99% data accuracy benchmark.
BR3	Accelerate on-site resolution timelines for deployed maintenance personnel.	Drive a 15% improvement in first-time asset resolution metrics.

2.3.2. Functional Requirements (FR)

The specifications below define the functional software capabilities required to support core business operations.

ID	Functional System Specification	Implementation Priority
FR1	The platform must retrieve and display complete customer profile records via an indexed Meter ID field within 2 seconds, without reliance on external platform integrations.	High Priority — Core capability required for real-time customer identification workflows.
FR2	The system must deliver a consolidated Service Console page layout designed for manual case entry during active telephone interactions.	High Priority — Required to remove dependency on legacy spreadsheet data entry.
FR3	The system must enable field technicians to view records and submit status updates through a secure mobile browser interface.	Medium Priority — Eliminates reliance on physical paper logs and informal WhatsApp-based dispatch.
FR4	The platform must trigger automated alert tracking for high-priority utility incidents that remain unassigned beyond 15 minutes.	Medium Priority — Reduces service delays during significant localized utility outages.

2.3.3. Non-Functional Requirements (NFR)

The parameters below establish the quality, safety, and performance boundaries governing the platform.

ID	Non-Functional Parameter	Technical Specification
NFR1	Cloud infrastructure availability must be sustained to support continuous field safety operations.	Maintain a minimum 99.9% platform uptime threshold throughout active business hours.
NFR2	Information security protocols must govern customer data visibility	Enforce a robust, multi-layered Role-Based Access Control (RBAC) framework across all user profiles.

	according to appropriate organizational levels.	
NFR3	User interface responsiveness must prioritize agent workspace efficiency.	Analytical dashboards and reports must load within a maximum of 5 seconds.

2.4 User Stories

2.4.1. User Story 1: Support Desk Live Call Processing

Background:

Elevated call volumes during service outages frequently cause the legacy Excel system to become unresponsive. Support agents require a unified interface to simultaneously identify callers and record case data, reducing overall wait times.

User Story:

- *As a Customer Support Agent,*
- *I want to retrieve customer records by Meter ID and log a service issue within a single Service Console page,*
- *So that I can handle calls efficiently without extending the customer's wait time.*

Acceptance Criteria:

- A Meter ID search must return the corresponding Account record within 2 seconds.
- Case entry forms must auto-fill the customer's phone number and location from the linked account record.

2.4.2. User Story 2: Field Maintenance Mobile Status Update

Background:

Informal WhatsApp-based dispatch contributes to a 15% location error rate, while paper-based logs are frequently misplaced. Field technicians require real-time data delivery and the ability to submit records directly from their mobile devices.

User Story:

- *As a Dispatched Field Technician,*

- *I want to receive immediate task assignments and submit resolution details directly from my mobile device,*
- *So that I can navigate to sites accurately and remove the dependency on paper-based logging.*

Acceptance Criteria:

- A mobile push notification must be delivered instantly upon assignment of a new work record.
- The Technical Resolution Notes field must be completed as a mandatory requirement before a record status can be set to "Resolved."

M3. Future State Process Design

3.1 Salesforce Process Design

3.1.1 Architectural Principles

a. Centralized Data Management

In order to move away from fragmented tracking sheets, Salesforce is designated as the authoritative single source of truth within the organization. The architecture leverages native data loading protocols to store both customer and meter records directly inside standard and custom entities, thereby removing any dependency on live external systems.

b. Declarative Automation Strategy

The solution is built around a declarative architecture philosophy — favoring configuration over custom code — by making full use of native Salesforce Flow and Assignment Rules. This strategy removes the need for manual dispatch steps while ensuring the platform remains stable over time and easier to govern.

c. Role-Based Security

System visibility is segmented according to each user group's operational scope and responsibilities, guaranteeing that data access is appropriately controlled across the organization.

3.1.2 Security and Governance Framework

To safeguard database records against unauthorized changes and support long-term scalability, the future-state design establishes a multi-layered data access strategy. This framework lays out the foundational governance pillars that will determine user visibility prior to the creation of explicit profile configurations.

a. Organization-Wide Defaults (OWD)

Private Baseline Restriction: Establishes a secure data boundary by restricting access to all records by default, preventing users from viewing client files they are not authorized to see.

b. Role Hierarchy Realignment

Vertical Reporting Roll-up: Automatically propagates record visibility upward through the management structure, enabling supervisors to maintain full operational oversight.

c. Functional Access Profiles

Role-Based Boundary Enforcements: Regulates feature access according to job responsibilities, drawing a clear line between administrative call desk functions and technical field-level entries.

3.2 Process Flow Diagrams

3.2.1 Future State TO-BE Process Flow

The redesigned future-state process introduces structured automation through Salesforce, aiming to replace manual spreadsheet management and accelerate information handoffs between roles.

- Step 1 (Intake): A customer contacts the support line to report an infrastructure disruption or service outage. The customer support agent picks up the call and launches the Salesforce Service Console.
- Step 2 (Synchronous Lookup): The agent enters the customer's unique identifier into the global search bar. Salesforce queries the indexed database and retrieves the associated Account, Contact, and asset information on a unified screen within 2 seconds (Resolution to Bottleneck B1).

- Step 3 (Logging): While still on the call, the agent confirms the customer’s details, records the issue parameters, and clicks save to create a new Case record, thereby concluding the customer-facing interaction.
- Step 4 (Asynchronous Dispatch): The act of saving the Case record automatically triggers a background Record-Triggered Flow in Salesforce. The system generates a related Field_Visit__c record and dispatches a targeted push notification directly to the designated technician’s mobile device (Resolution to Bottleneck B2).
- Step 5 (Execution): Upon receiving the mobile alert, the field technician proceeds to the specified coordinates, carries out the required physical repairs, and marks the status field as “Resolved” within the Salesforce Mobile App.
- Step 6 (Real-Time Closeout): The mobile status update activates a backend automation flow that transfers the technical notes, automatically sets the parent Case to “Closed”, and instantly refreshes the executive KPI dashboards (Resolution to Bottleneck B3).

3.3 Gap Analysis

The table below compares the manual shortcomings identified in the legacy environment against the structured solutions developed for the Salesforce configuration.

Legacy Identified Gap (AS-IS)	Salesforce Optimized Solution (TO-BE)	Target Enterprise Business Value
Manual lookups in spreadsheets consumed up to 5 minutes per customer interaction.	Customer and asset records are natively structured with indexed lookup fields.	Reduces search time to under 2 seconds per call, cutting abandonment rates.
Case details were manually typed into WhatsApp or communicated over voice calls.	Automated assignment logic with Record-Triggered custom object creation handles routing.	Eliminates manual dispatch coordination, closing a 15% human error gap.
Field personnel logged technical parameters on handwritten paper forms.	Technicians capture resolution data natively through standard mobile page layouts.	Prevents information loss from field operations and removes paper-based overhead.

Job record updates were delayed 2 to 4 hours before reaching management.	Automatic case roll-up logic refreshes dashboards the moment a record is changed.	Gives leadership immediate, clean visibility into real-time operational KPIs.
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M4. Data Model Design

4.1 Object Definition

To establish a scalable architecture that accurately represents physical infrastructure assets while preventing duplicate records, a balanced combination of standard and custom objects has been structured below.

4.1.1 Standard Objects

- **Account:** Represents the primary consumer entity or household receiving active utility distribution services from GreenGrid.
- **Contact:** Manages the detailed personal identification information, phone numbers, and communication logs for individual billing representatives.
- **Case:** Serves as the central transactional ticket to log, prioritize, track, and resolve active customer service telephone complaints.

4.1.2 Custom Objects

- **Utility_Asset__c:** Tracks the physical utility meter hardware grid installation asset. This object stores the alphanumeric identifier field natively within Salesforce to guarantee high-performance search metrics.
- **Field_Visit__c:** Tracks the history of remote on-site dispatches, travel variables, repair assignments, and technical resolution details handled by the field crew.

4.2 Field Definition

4.2.1 Standard Object

To accommodate the specialized customer service workflows needed for utility infrastructure management, the standard Salesforce Case object was customized. Instead of introducing a redundant custom entity, the out-of-the-box **Type** field

metadata was expanded with industry-specific picklist values. This classifies incoming support tickets efficiently while leveraging standard platform optimization.

Standard Object: Case

Field Label	API Name	Data Type	Description and Validation Rules
Case Type	Type	Picklist	* Outage: Used to classify distribution grid failures. * Maintenance: Used for scheduling routine technical grid adjustments and asset tracking updates to mitigate active failure risks.

4.2.2 Custom Object

The tables below outline the critical technical fields needed to execute business logic, enforce data validation, and meet the requirement of under 2-second search performance metrics.

Custom Object: Utility_Asset__c

Field Label	API Name	Data Type	Description and Validation Rules
Asset Name	Name	Auto-Number	Automatically generated system identifier (Format: MET-{0000}).
Meter ID	Meter_ID__c	Text (Length: 50)	Unique, indexed identifier field used by agents for rapid customer lookups. Set as Unique and External ID .
Asset Status	Asset_Status__c	Picklist	Tracks operational state. Values: <i>Active</i> , <i>Inactive</i> , <i>Suspended</i> .

Installation Address	Installation_Address__c	Text Area (Length: 255)	Holds the exact physical geographic address where the meter asset is deployed.
Customer Account	Customer_Account__c	Lookup (Account)	Establishes a flexible relationship linking the physical hardware asset to a parent customer account.

Custom Object: Field_Visit__c

Field Label	API Name	Data Type	Description and Validation Rules
Visit Number	Name	Auto-Number	Automatically generated record identifier (Format: VIS-{0000}).
Parent Case	Parent_Case__c	Lookup (Case)	Connects the dispatch log to its original support ticket. <i>Validation Rule:</i> Clear value on delete or restrict deletion to preserve history.
Repair Status	Repair_Status__c	Picklist	Mandatory operational tracking field. Values: <i>Assigned, In Progress, Resolved, Failed.</i>
Technician Notes	Technician_Notes__c	Long Text Area (32,000)	Long text field where technicians type site findings directly into the mobile layout.
Resolution Timestamp	Resolution_Timestamp__c	Date/Time	Automatically populated by system flows when status shifts to <i>Resolved</i> to track KPI dashboards.

4.3 Relationship Mapping

This section details the relational structure of the database schema, illustrating how standard and custom entities link together within the GreenGrid Utilities Salesforce environment. By applying specific database constraints, this architecture protects regulatory historical compliance data and enforces reference integrity.

The matrix below defines the relationship types, parent-child structures, and exact transactional system behaviors applied during deletion scenarios.

Parent Object	Child Object	Relationship Type	Data Behavior and Deletion Constraint
Account	Contact	Standard Lookup	Clear Value on Delete: If an Account record is deleted, related Contact records remain in the database with their lookup reference field cleared.
Account	Utility_Asset__c	Custom Lookup	Clear Value on Change: If a customer account is modified or closed, the physical meter asset record remains un-deleted and can be re-assigned to the incoming occupant account.
Account	Case	Standard Lookup	Restricted Deletion: The system blocks the deletion of an Account record if active or historical Case records are attached to it, securing critical customer audit logs.
Utility_Asset__c	Case	Custom Lookup	Clear Value on Change: If hardware assets are replaced or updated, historical service ticket logs remain safely intact for utility grid pattern analytics.

Case	Field_Visit__c	Custom Lookup	Restricted Deletion: The system prevents the manual deletion of an active Field Visit record if it contains unresolved technical notes, ensuring audit compliance for field operations.
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- **Account to Contact Relationship:** Formed as a standard Lookup connection. This allows a primary customer household account to act as the main hub for multiple individual contact records (such as secondary residents or property managers) while keeping data independent.
- **Account to Utility_Asset__c Relationship:** Designed as a custom Lookup connection. This choice uncouples the physical utility meter hardware from changing human residents occupying the property. When property ownership changes, the asset record remains bound to its geographic installation address, and the lookup field is simply updated to the new Account.
- **Account to Case Relationship:** Configured as a standard Lookup connection governed by a strict Restricted Deletion rule. This safeguard ensures that legacy support interactions, outage history, and resolution timelines remain archived for corporate SLA reporting, even if a user tries to delete the primary customer account file.
- **Utility_Asset__c to Case Relationship:** Set up as a custom Lookup connection. Linking the ticket directly to the hardware asset allows helpdesk agents to instantly check the asset's mechanical repair history during an active call, speeding up root-cause diagnosis.
- **Case to Field_Visit__c Relationship:** Set up as a custom Master-Detail connection. This tight structural bond ensures that field workers can only generate a field visit if it is tied to an active, authorized parent case. Additionally, the child record automatically inherits the security visibility rules, owner assignments, and sharing permissions defined on the parent Case.

4.4 Salesforce Implementation and Configuration

Step 1: Creating a Developer Account

1. Navigate to the Salesforce Developer Edition Sign-up page.

2. Fill out the registration form with your details (First Name, Last Name, Email, Role, Company, Country, and Postal Code).

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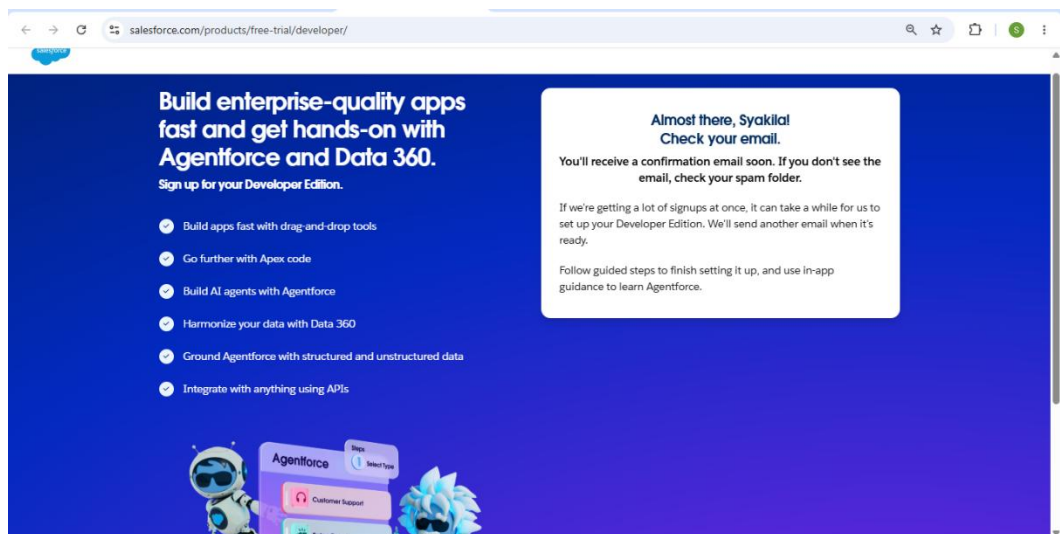
First name: Syakila ✓ Last name: Aini ✓
Job title: Student ✓ Company: SmartBridge ✓
Country/region: Indonesia ✓ Work email: syakilaaini01@gmail.com ✓

☒ I agree to the [Main Services Agreement - Developer Services and Salesforce Program Agreement](#)
☒ Yes, I would like to receive marketing communications regarding Salesforce products, services, and events. I can unsubscribe at any time.

Your free trial may be provisioned on or migrated to Hyperforce, Salesforce's public cloud infrastructure.
I acknowledge, as described in the Developer Documentation: (1) the Developer Edition includes autonomous and other generative AI features; and (2) Salesforce may limit use of those features and the org, and may terminate any org that has been inactive for 45 days.
By submitting this form, you confirm that you agree to the processing of your personal data, including transfer and processing outside of Indonesia, as described in the [Privacy Statement](#).

Sign me up

3. Click the "Sign me up" button and wait for the confirmation email to arrive in your inbox.



Step 2: Account Activation

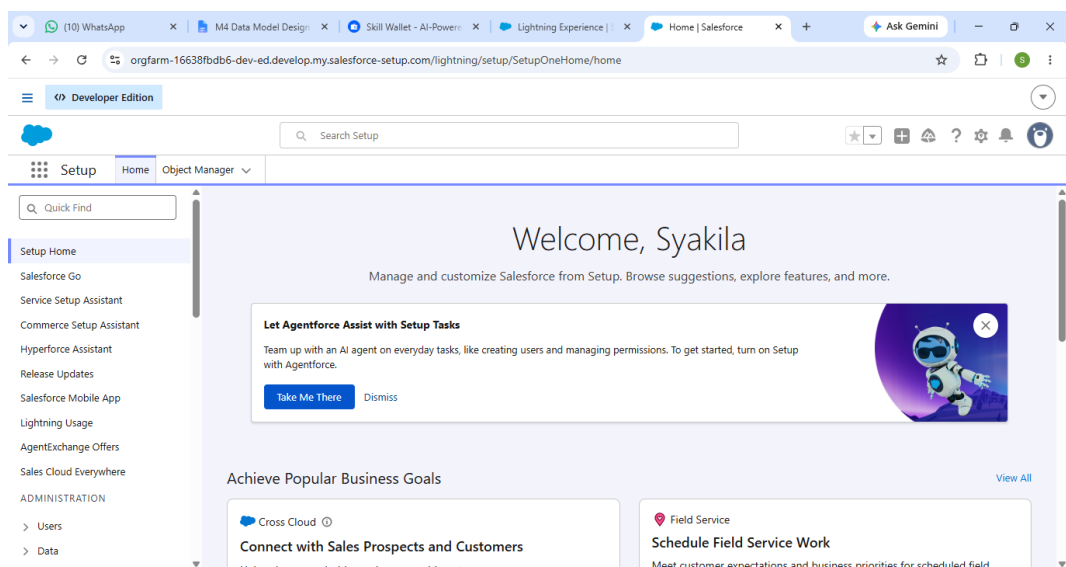
1. Access the inbox of the email address provided during sign-up.
2. Open the verification email and click the **"Reset Password"** button to activate your account (this email may take 2 to 10 minutes to arrive).
3. Set up your new password, choose a security question, provide an answer, and click **"Change Password"**.

4. You will be automatically redirected to the Salesforce Setup home screen.

Step 3: Custom Object Creation for Utility Asset

The primary hardware tracking table was established by generating a new custom object inside the Object Manager tool. Follow these actions:

1. From the Setup home page, click on the **Object Manager** tab.
2. Click **Create** in the upper right-hand corner, then select **Custom Object**.



3. Define the object configuration parameters as follows:
 - **Label:** Utility Asset
 - **Plural Label:** Utility Assets
 - **Object Name (API Name):** Utility_Asset
 - **Record Name:** Asset Name
 - **Data Type:** Auto-Number (Display Format: MET-{0000}, Starting Number: 1)
4. Under **Optional Features**, check the following boxes:
 - *Allow Reports*
 - *Allow Track Field History*

5. Under **Search Status**, check **Allow Search**.

6. Click **Save**.

The screenshot shows the Salesforce 'New Custom Object' setup page for a custom object named 'Utility Asset'. The page is titled 'New Custom Object' and includes a search bar at the top. The main content area is divided into several sections:

- Custom Object Definition Edit:** This section contains fields for 'Label' (Utility Asset), 'Plural Label' (Utility Assets), and 'Object Name' (Utility__Asset). It also has a 'Description' field and a 'Context-Sensitive Help Setting' section.
- Enter Record Name Label and Format:** This section includes a 'Record Name' field (Account Name) and a 'Warning' message about high volume of records.
- Optional Features:** This section includes checkboxes for 'Allow Reports', 'Allow Activities', 'Track Field History', 'Allow in Chatter Groups', and 'Enable Licensing'.
- Object Classification:** This section includes checkboxes for 'Allow Sharing', 'Allow Bulk API Access', and 'Allow Streaming API Access'.
- Deployment Status:** This section includes radio buttons for 'In Development' and 'Deployed'.
- Search Status:** This section includes a checkbox for 'Allow Search'.
- Object Creation Options:** This section includes checkboxes for 'Add Notes and Attachments related list to default page layout' and 'Launch New Custom Tab Wizard after creating this custom object'.

The page concludes with 'Save', 'Save & New', and 'Cancel' buttons.

Step 4: Custom Field Creation for Utility Asset

Specific technical metadata fields were configured under the **Fields & Relationships** section of the new Utility Asset object to capture operational data:

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Asset Name	Name	Auto Number		✓
Asset Status	Asset_Status__c	Picklist		
Created By	CreatedBy	Lookup(User)		
Customer Account	Customer_Account__c	Lookup(Account)		✓
Installation Address	Installation_Address__c	Text Area(255)		
Last Modified By	LastModifiedBy	Lookup(User)		
Meter ID	Meter_ID__c	Text(50)		✓
Owner	OwnerId	Lookup(User:Group)		✓

- **Meter ID (Meter_ID__c):** Created as a Text field with a character length of 50. To satisfy the high-priority business requirement of sub-2-second data retrieval speeds, this attribute was explicitly enabled as both a **Unique** record identifier and an **Indexed External ID**.
- **Asset Status (Asset_Status__c):** Configured as a Picklist data type containing explicit system values: *Active*, *Inactive*, and *Suspended*.
- **Installation Address (Installation_Address__c):** Configured as a standard Text Area field capped at 255 characters to efficiently store geographic location strings.
- **Customer Account (Customer_Account__c):** Established as a flexible Lookup Relationship pointing directly to the standard Account entity to connect assets to active consumers.

Step 5: Custom Object Creation for Field Visit

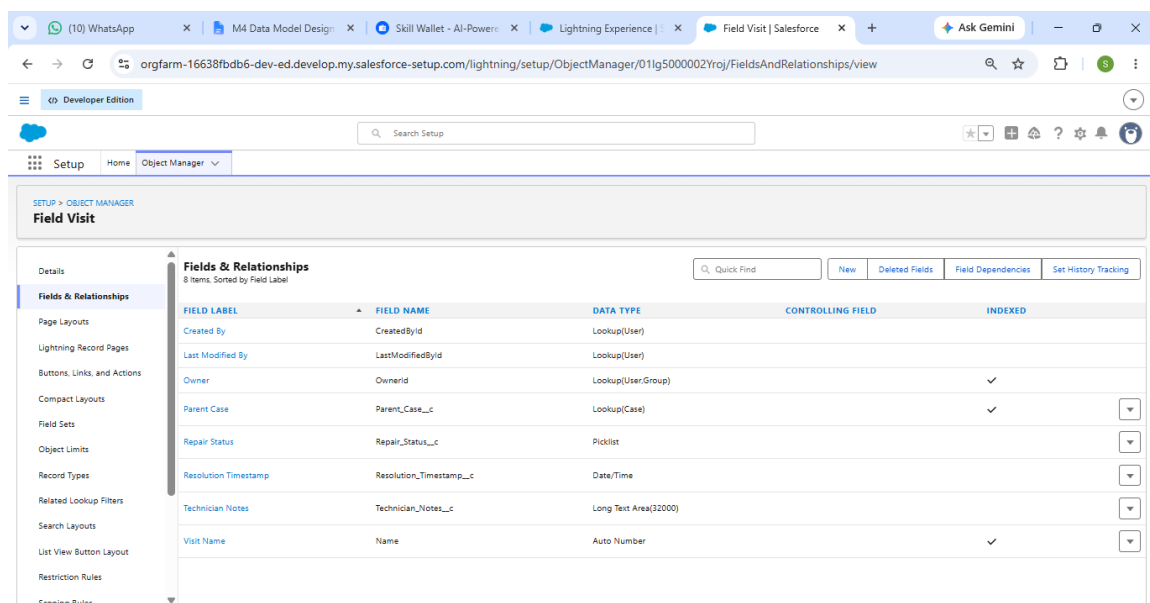
To move the technical field crew dispatch processes away from legacy paper-based tracking methods, a secondary custom object was created inside the Object Manager layout.

- **Label:** Field Visit
- **Plural Label:** Field Visits
- **Object Name (API Name):** Field_Visit

- **Record Name:** Visit Number
- **Data Type:** Auto-Number (Display Format: VIS-{0000}, Starting Number: 1)
- *Follow the identical creation steps outlined in Step 3.*

Step 6: Custom Field Creation for Field Visit

To maintain operational tracking flexibility and support offline mobile user requirements, data properties were added to the Field Visit object framework:



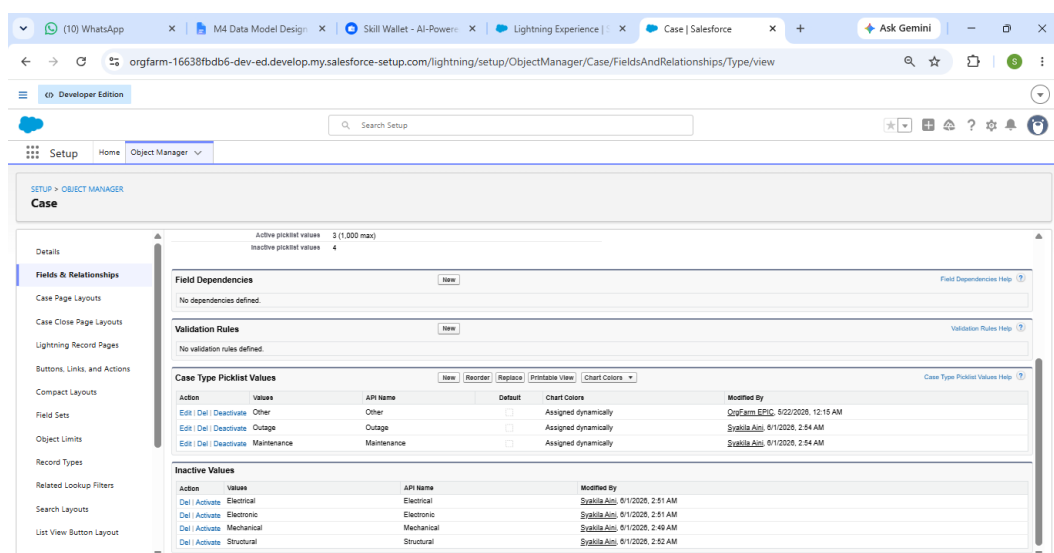
FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedBy	Lookup(User)		
Last Modified By	LastModifiedBy	Lookup(User)		
Owner	OwnerId	Lookup(User/Group)		✓
Parent Case	Parent_Case__c	Lookup(Case)		✓
Repair Status	Repair_Status__c	Picklist		
Resolution Timestamp	Resolution_Timestamp__c	Date/Time		
Technician Notes	Technician_Notes__c	Long Text Area(32000)		
Visit Name	Name	Auto Number		✓

- **Parent Case (Parent_Case__c):** Set up as a standard Lookup Relationship pointing directly to the Case entity. This structural choice ensures that field workers can maintain independent record ownership on their mobile devices while staying connected to the root support ticket.
- **Repair Status (Repair_Status__c):** Configured as a Picklist data type with workflow status values: *Assigned, In Progress, Resolved, and Failed*.
- **Technician Notes (Technician_Notes__c):** Configured as a Long Text Area (32,000 characters) to allow detailed mobile technical descriptions.
- **Resolution Timestamp (Resolution_Timestamp__c):** Configured as a Date/Time field to log the exact completion moment for operational efficiency metrics.

Step 7: Standard Object Metadata and Page Layout Customization (Case)

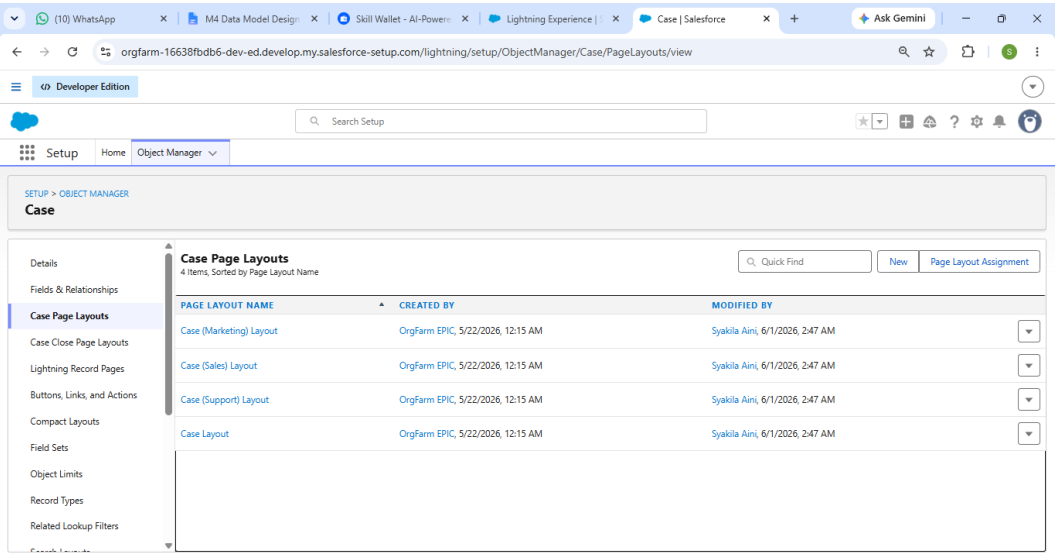
Before verifying the final system schema, the standard Case object infrastructure is customized to recognize utility-specific support categories and clean up the user interface.

1. From the Setup page, click on **Object Manager** in the top menu bar.
2. Scroll down and click on the standard **Case** object.
3. Choose **Fields & Relationships**, then click the **New** button.
4. Select **Lookup Relationship** as the Data Type and click **Next**.
5. In the **Related To** dropdown menu, select **Utility Asset** and click **Next**.
6. Set the Field Label to Utility Asset (the Field Name will auto-populate as Utility_Asset) and click **Next**.
7. Advance through the field-level security visibility screens, leave the field added to the layouts, and click **Save**.
8. Remaining under Case Fields & Relationships, click on the standard field **Type** (Picklist).
9. Scroll down to the **Case Type Picklist Values** related list and click **New** to add Outage and Maintenance on separate lines. Click **Save**.

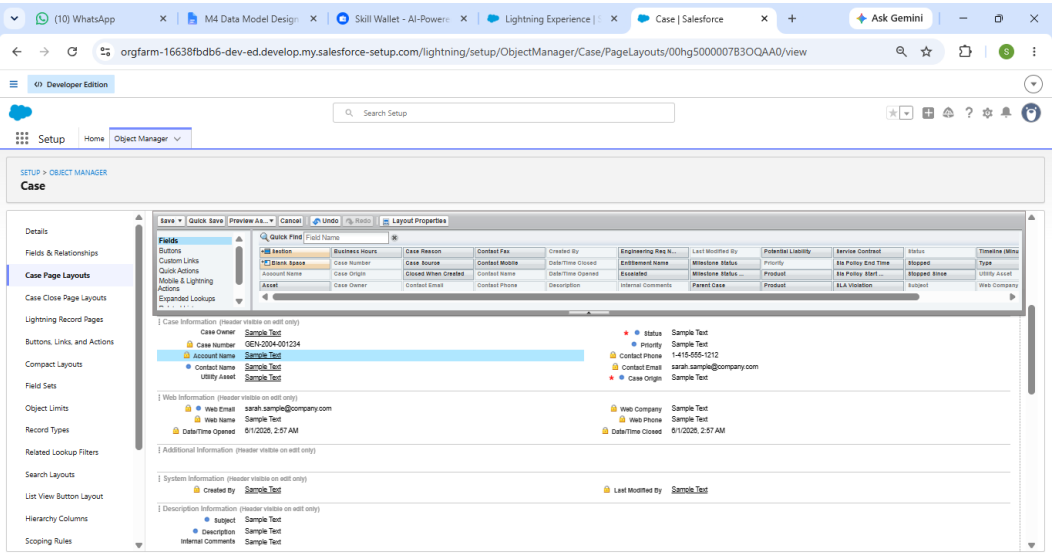


10. To clean up the interface and enforce strict data standardization, locate the standard default values (*Mechanical*, *Electrical*, *Electronic*, *Structural*, and *Other*) within the active values table, and click **Del / Deactivate** next to each entry.

11. From the left navigation menu of the Case object, click on **Case Page Layouts** and select **Case Layout**.



12. Streamline the Case Page Layout by removing irrelevant standard fields and positioning the custom **Utility Asset** and updated **Type** fields prominently.

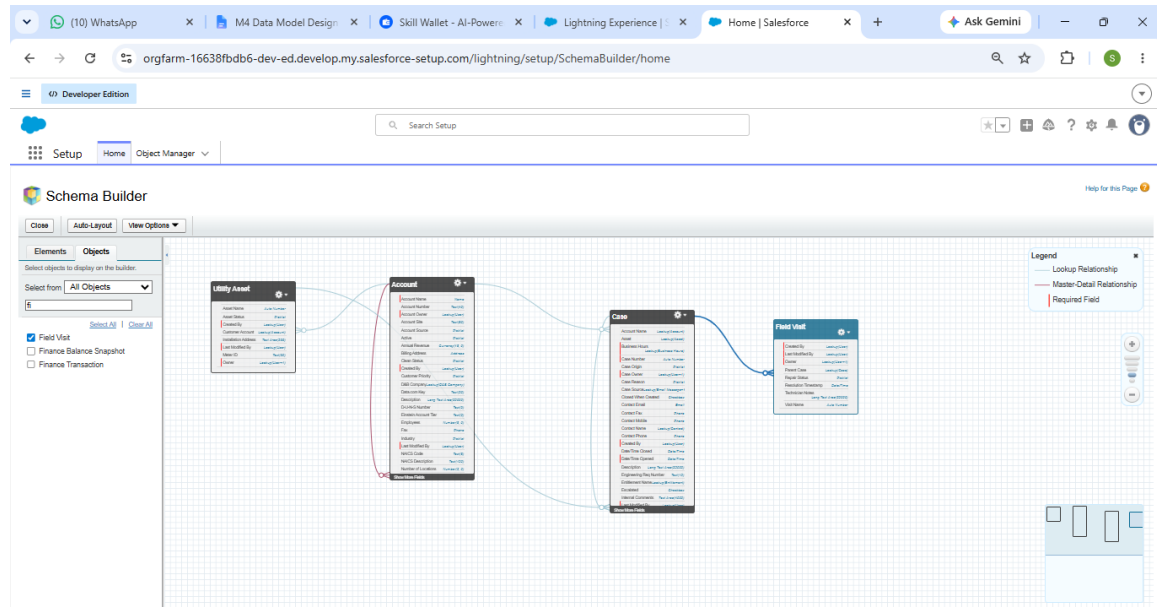


13. Click **Save** on the upper left-hand layout palette menu.

Step 8: System Schema Integrity Verification

The native Salesforce **Schema Builder** utility was utilized to run a database layout audit. By isolating the standard Account and customized Case entities alongside the custom

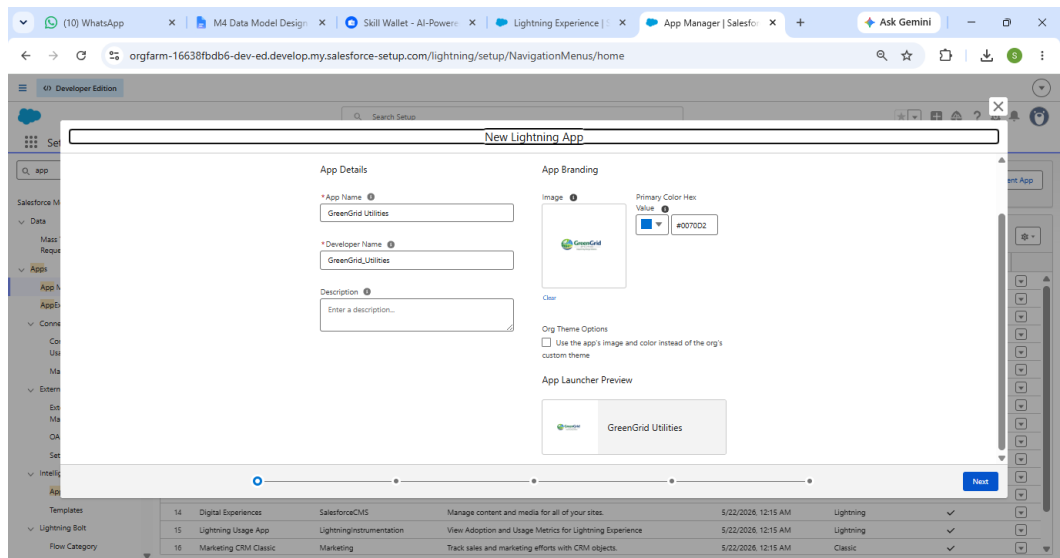
Utility Asset and Field Visit objects, the data model structure was visually checked to ensure reference integrity before entering the security implementation phase.



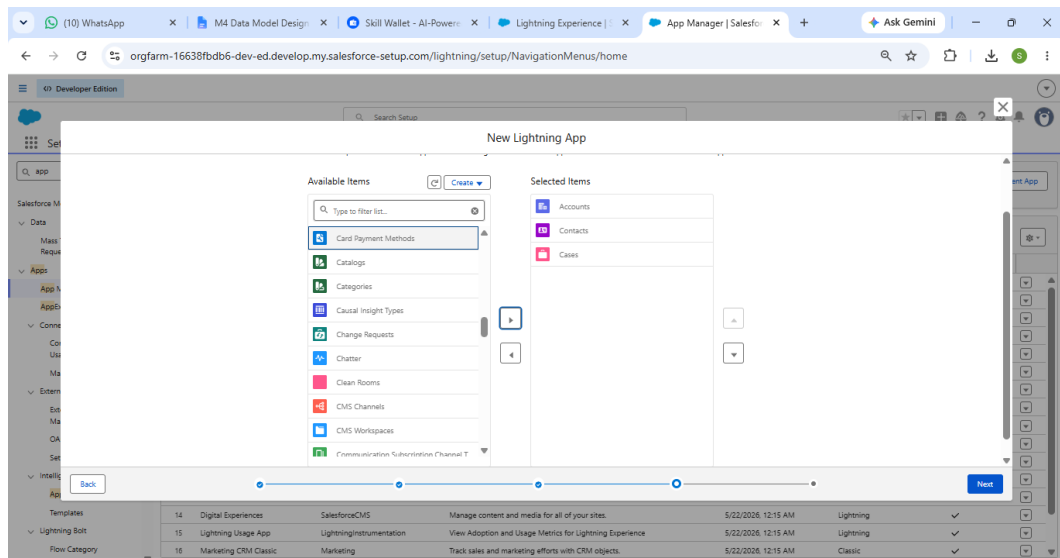
Step 9: Custom Lightning App Creation

To establish a dedicated, branded workspace layout for end-users, a custom Lightning Application container was provisioned within the platform App Manager.

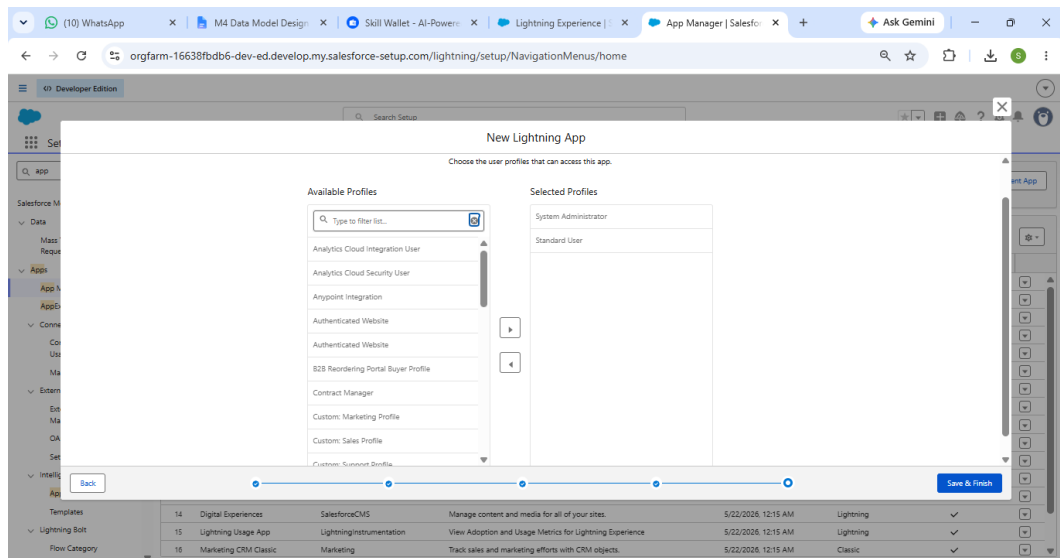
1. Navigate to **Setup** → type **App Manager** in the Quick Find bar, click on **App Manager**.
2. Click the **New Lightning App** button located in the upper right-hand corner.
3. Fill out the App Name as GreenGrid Utilities, upload branding assets if needed, and click **Next**.



4. Advance through the **App Options** and **Utility Items** parameters by keeping the standard system default settings.
5. On the **Navigation Items** panel, select standard entities **Accounts**, **Contacts**, and **Cases** from the available list and move them to the **Selected Items** box to form the base configuration baseline.



6. On the **User Profiles** assignment step, scroll down and assign active visibility access to the **System Administrator** profile along with core operational user profiles.

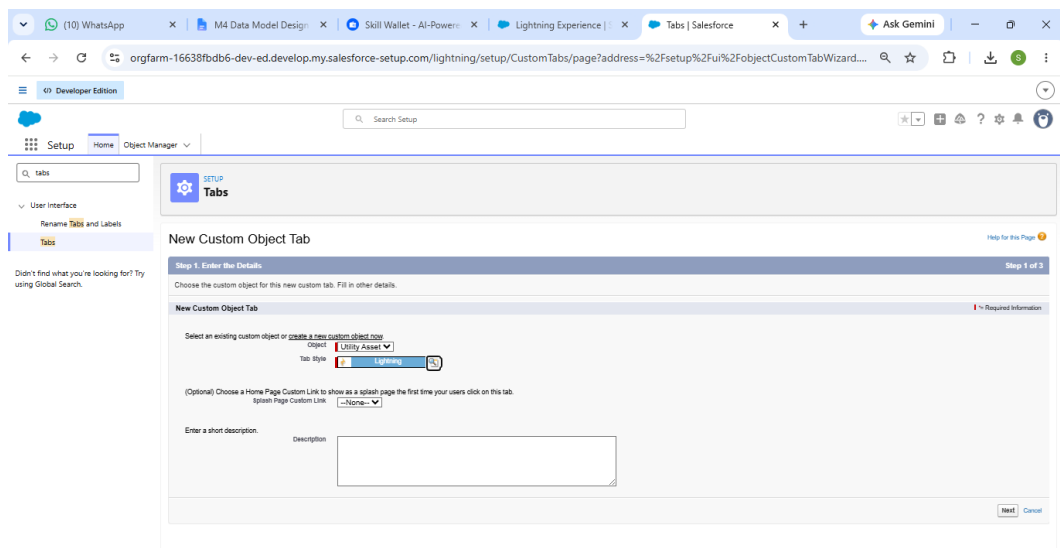


7. Click **Save & Finish**.

Step 10: Custom Object Tab Provisioning (Utility Asset)

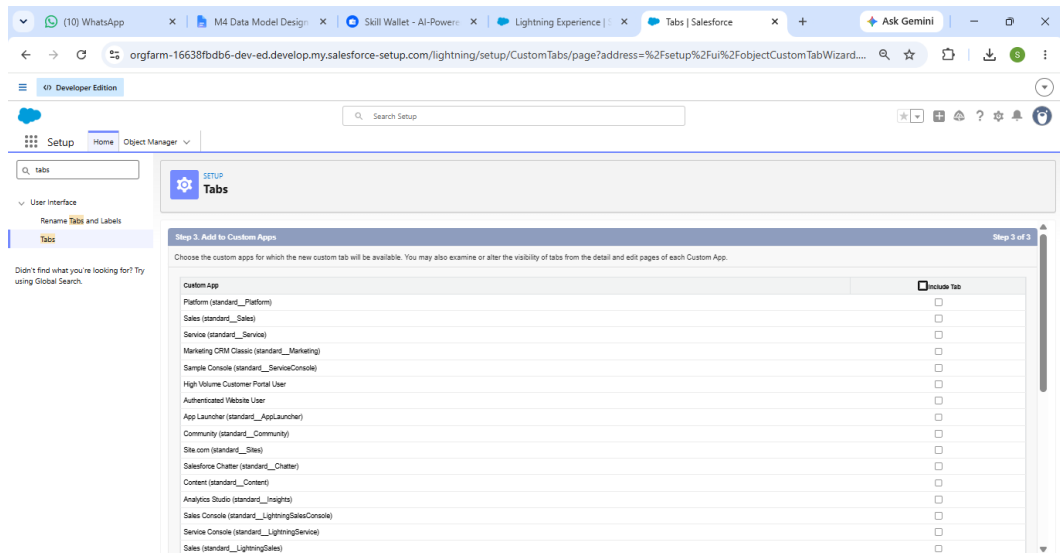
To make the newly deployed custom tables visible and accessible to system end-users within the navigation interface, standalone user interface tabs were provisioned.

1. Navigate to **Setup**, type **Tabs** in the Quick Find bar, click on **Tabs**.
2. Under the **Custom Object Tabs** related list, click the **New** button.



3. Select **Utility Asset** from the Object dropdown list and assign a distinct visual **Tab Style**. Click **Next**.

4. Leave the profile visibility settings at their default values and click **Next** again.
5. On the **Add to Custom Apps** step, uncheck the "Include Tab" global header box to remove it from irrelevant background applications, but ensure the checkbox remains explicitly checked for the **GreenGrid Utilities App**.

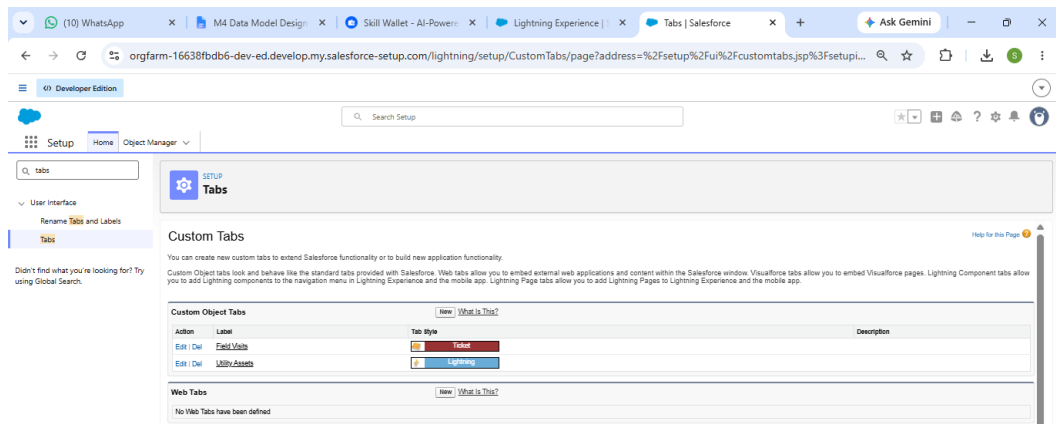


6. Ensure the checkbox for **"Append tab to users' existing personal customizations"** is checked at the bottom of the page layout.
7. Click **Save**.

Step 11: Custom Object Tab Provisioning (Field Visit)

Following the successful deployment of the primary infrastructure layout, navigation tabs for the remaining operational objects were built using the same method as Step 10 to complete the end-user navigation framework.

1. Return to the **Tabs** menu and click **New** under the Custom Object Tabs panel.
2. Select **Field Visit** from the Object dropdown list and assign a distinct visual **Tab Style**.
3. Advance to the app routing parameters screen, exclude the tab from general background profiles, and explicitly enable it for the **GreenGrid Utilities App** navigation bar.



4. Verify the personal customization append rule is active and click **Save**.

M5. Security Model and Access Control

5.1 Role Hierarchy Design

The GreenGrid Utilities role hierarchy is built directly around the High-Power and High-Interest stakeholders identified during the Milestone 1 phase.

The main goal of this architecture is to dynamically handle vertical data visibility. When Organization-Wide Defaults (OWD) are set to restrictive levels, the role hierarchy automatically opens up record visibility upward through the management line. This allows executive leadership to see real-time data and metrics without needing manual sharing rules.

To maintain consistency across all project assets, the Salesforce roles map exactly to the team definitions established in the project charter and process swimlanes:

- **Executive Management (Top Tier):** Positioned at the very top of the hierarchy. This role has complete, unrestricted access to all underlying Cases, asset operations, and technician metrics to feed their live management dashboards.
- **Customer Support Team (Baseline Tier):** Staff in this role manage incoming calls and create new records. They operate at a foundational level and are laterally blocked from viewing field technician records to keep their workspace focused.
- **Field Technicians (Baseline Tier):** Operating as a peer level to the support desk, this role is for remote, on-site workers. They own individual records directly on their mobile devices, and their data remains isolated from the support team's view.

5.2 Permission Sets Definition

Following the principle of **Least Privilege Access**, base user profiles are restricted to basic operational views. Advanced data management features are granted dynamically using Permission Sets. Since the system uses flexible Lookup Relationships instead of rigid Master-Detail restrictions, Permission Sets are used to give field technicians independent record ownership and editing capabilities.

The system uses two dedicated permission set extensions:

5.2.1 Field Technician Mobile Access (Field_Technician_Mobile_Access)

This is assigned only to users in the Field Technicians role to allow real-time mobile validation and data entry directly from service locations.

Target Object	Object-Level Permission	Field-Level Access Parameter
Account	Read-Only	Limited to customer identity and billing location strings.
Case	Read-Only	Limited to historical descriptions and priority fields for context.
Utility_Asset__c	Read-Only	Read-Only view of equipment serial numbers and indexing keys.
Field_Visit__c	Read, Create, Update	Full editing rights for Repair_Status__c and Technician Notes fields.

5.2.2 Customer Support Escalation Permissions (Support_Desk_Escalation_Access)

This container is given to senior coordinators on the Customer Support Team who need elevated administrative control during critical grid outages.

Target Object	Object-Level Permission	Field-Level Access Parameter
Case	Read, Edit, Delete	Allows changing record ownership and updating escalation priority metrics.

Field_Visit__c	Read, Edit	Allows balancing technical work distribution across queues.
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5.3 Sharing Rules Configuration

To prevent duplicate data and protect underlying utility parameters, standard baseline sharing settings are locked down. Automated sharing rules are used purely as exceptions to open up data lines based on business needs.

5.3.1 Organization-Wide Defaults (OWD) Baseline

Data security settings for standard and custom objects are configured as follows to safeguard information:

Object Name	API Name	OWD Setting	Protection Target
Account	Account	Private	Stops cross-regional customer browsing and duplication.
Contact	Contact	Private	Protects primary phone numbers and communication histories.
Case	Case	Private	Restricts active phone logs to the assigned support agent's view.
Utility Asset	Utility_Asset__c	Private	Protects master equipment serial numbers and grid asset nodes.
Field Visit	Field_Visit__c	Private	Restricts technical task details to assigned mobile operator queues.

5.3.2 Criteria-Based Automated Sharing Rule

To meet the business goal of reducing response times during emergencies, an automated sharing exception rule has been set up in the platform.

- **Rule Developer Name:** High_Priority_Case_Sharing
- **Object Type:** Case
- **Rule Type:** Based on criteria

- **Field Criteria Parameter:** Priority EQUALS High AND Status EQUALS New OR Working
- **System Action Sharing Target:** Automatically shares matching Case records with the **Executive Management** public role group with **Read/Write** permissions.
- **Business Justification:** This exception allows major grid infrastructure issues to bypass base data restrictions, giving executive leaders immediate visibility into critical problems.

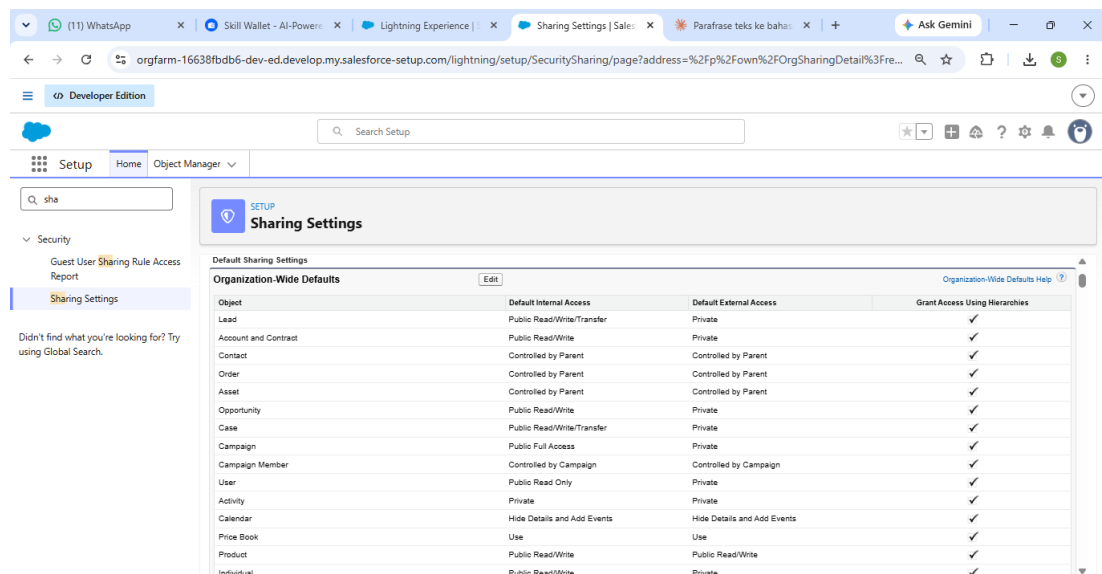
5.4 Salesforce Implementation and Configuration

This section provides the step-by-step instructions for manually configuring this data control model in a Salesforce Developer Edition environment.

Step 1: Organization-Wide Defaults (OWD) Lockdown

Secure the system data access by changing the sharing configuration inside the Sharing Settings utility.

1. Go to **Setup, Security, Sharing Settings**.
2. Click **Edit** on the Organization-Wide Defaults section.
3. Change the **Default Internal Access** column to **Private** for the following objects: Account, Contact, Case, Utility_Asset__c, and Field_Visit__c.
4. Click **Save & OK**, then wait for the system confirmation email regarding data recalculation.



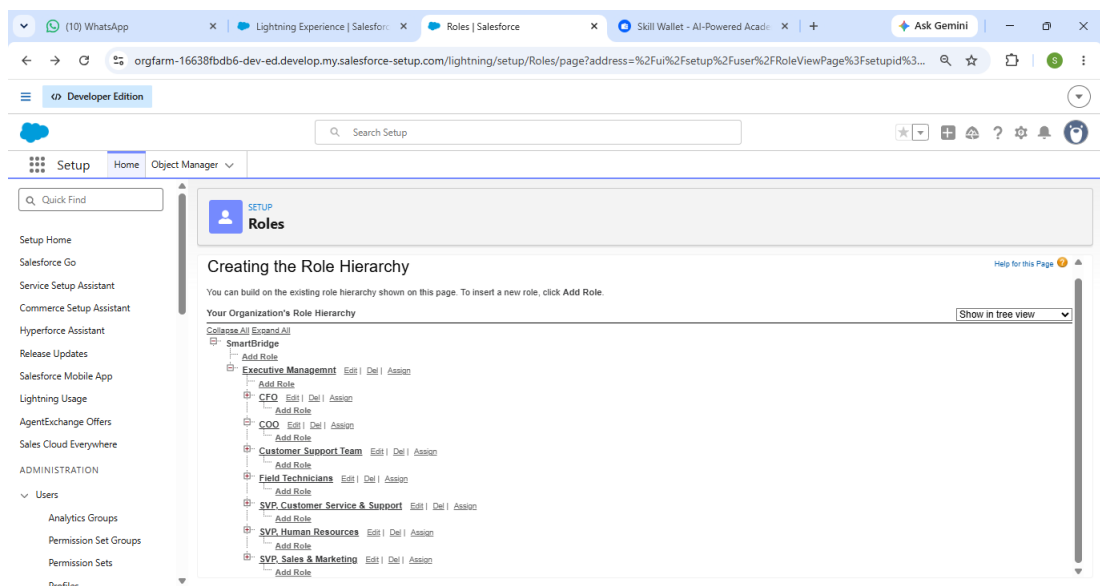
The screenshot shows the Salesforce Sharing Settings page. The left sidebar contains the navigation menu with 'Setup', 'Home', and 'Object Manager'. The main content area is titled 'Sharing Settings' and shows the 'Organization-Wide Defaults' section. A table lists various objects with their 'Default Internal Access', 'Default External Access', and 'Grant Access Using Hierarchies' settings. The 'Default Internal Access' column is being edited to 'Private' for several objects.

Object	Default Internal Access	Default External Access	Grant Access Using Hierarchies
Lead	Public Read/Write/Transfer	Private	✓
Account and Contract	Public Read/Write	Private	✓
Contact	Controlled by Parent	Controlled by Parent	✓
Order	Controlled by Parent	Controlled by Parent	✓
Asset	Controlled by Parent	Controlled by Parent	✓
Opportunity	Public Read/Write	Private	✓
Case	Public Read/Write/Transfer	Private	✓
Campaign	Public Full Access	Private	✓
Campaign Member	Controlled by Campaign	Controlled by Campaign	✓
User	Public Read Only	Private	✓
Activity	Private	Private	✓
Calendar	Hide Details and Add Events	Hide Details and Add Events	✓
Price Book	Use	Use	✓
Product	Public Read/Write	Public Read/Write	✓
Individual	Public Read/Write	Private	✓

Step 2: Role Hierarchy Structure Configuration

Set up the corporate organizational tree to control data visibility and automatic management roll-ups.

1. Go to **Setup, Users, Roles**.
2. Choose the corporate hierarchy view and click **Set Up Roles**.
3. Find the standard, default **CEO** role located right below the main company root node, and click **Edit**.
4. Rename the **Role Label** to Executive Management (the Role Name will automatically become Executive_Management). Click **Save**.
5. Click **Add Role** directly underneath the new Executive Management node to create the first sub-branch named Customer Support Team. Click **Save**.
6. Go back to the Executive Management node, click **Add Role** again to create the parallel sub-branch named Field Technicians, and click **Save**.



Step 3: Custom Permission Set Construction

Create a specialized permission container to give field operations extra object and field-level access without changing their base profiles.

1. Go to **Setup, Users, Permission Sets** and click **New**.
2. Label the permission set Field Technician Mobile Access (API Name: Field_Technician_Mobile_Access) and click **Save**.

Permission Set
Create

Enter permission set information

Label: Field Technician Mobile Access

API Name: Field_Technician_Mobile_Access

Description:

Session Activation Required: ☐

Select the type of users who will use this permission set

Who will use this permission set?

- Choose -None- if you plan to assign this permission set to multiple users with different user and permission set licenses.
- Choose a specific user license if you want users with only one license type to use this permission set.
- Choose a specific permission set license if you want this permission set license auto-assigned with the permission set.

Not sure what a permission set license is? [Learn more here.](#)

License: ...None...

3. Open **Object Settings**, scroll down, select **Field Visits**, and click **Edit**.
4. Check the boxes for **Read**, **Create**, and **Edit**. Under **Field Permissions**, check **Edit**

Permission Sets

Object Permissions

Permission Name	Enabled
Read	☒
Create	☒
Edit	☒
Delete	☐
View All Records	☐
Modify All Records	☐
View All Fields	☐

Field Permissions

Field Name	Field API Name	Read Access	Edit Access
Created By	CreatedById	☐	☐
Last Modified By	LastModifiedById	☐	☐
Owner	OwnerId	☐	☐
Parent Case	Parent_Case__c	☐	☐
Repair Status	Repair_Status__c	☒	☒
Resolution Timestamp	Resolution_Timestamp__c	☐	☐
Technician Notes	Technician_Notes__c	☒	☒
Visit Name	Name	☐	☐

5. **Access** for both **Repair Status** and **Technician Notes**.
6. Click **Save**.

Step 4: Automated Case Sharing Rule Deployment

Activate the automated data sharing exception to push high-priority cases to management queues.

1. Scroll down the main **Sharing Settings** page to the **Case Sharing Rules** related list and click **New**.
2. Label the rule **High Priority Case Sharing**. Set the **Rule Type** to **Based on criteria**.

3. Set the Criteria to: Priority EQUALS High.
4. Set **Share With** to: **Role, Executive Management**.
5. Set **Case Access** level to: **Read/Write**.

The screenshot shows the Salesforce Setup page for a Case Sharing Rule. The page is titled "Sharing Settings" and "Case Sharing Rule". It includes a search bar and a navigation menu on the left. The main content area is divided into five steps:

- Step 1: Rule Name**: Label is "High Priority Case Sharing", Rule Name is "High_Priority_Case_Sharing", and Description is empty.
- Step 2: Select your rule type**: Rule Type is "Based on criteria" (selected).
- Step 3: Select which records to be shared**: Criteria table with columns Field, Operator, and Value. The first row is "Priority" equals "High". There are three empty rows below it. The "AND" button is selected.
- Additional Options**: "Include records owned by users who can't have an assigned role" is checked.
- Step 4: Select the users to share with**: Share with is "Roles" and "Executive Management".
- Step 5: Select the level of access for the users**: Case Access is "Read/Write".

At the bottom, there are "Save" and "Cancel" buttons.

6. Click **Save** (and click **OK** if a confirmation pop-up appears).

Step 5: Operational User Provisioning and Role Assignment

Create standard test users to verify that the Private OWD and vertical Role Hierarchy work correctly under live conditions.

1. Go to **Setup, Users, Users** and click the **New User** button.
2. Create the test user for the support desk environment:
 - **First Name:** Support | **Last Name:** Agent
 - **Email / Username:** Use a distinct format (e.g., support_@greengrid.dev).
 - **Role:** Assign to the Customer Support Team node.
 - **User License:** Salesforce
 - **Profile:** Standard User

New User

Salesforce now sends emails only from verified domains. To avoid delivery failures, verify your email domains or enable your substitute email-sending domain. [Learn More](#)

User Edit [Cancel] [Save] [Save & New]

General Information

First Name	Support	Role	Customer Support Team
Last Name	Agent	User License	Salesforce
Alias	sagen	Profile	Standard User
Email	syakilaan61+cs@gmail.co	Active	☒
Username	support_@greengrid.dev	Marketing User	☐
Nickname	User178038113264858080	Offline User	☐
Title		Knowledge User	☐
Company		Flow User	☐
Department		Service Cloud User	☐
Division		Site.com Contributor User	☐
		Site.com Publisher User	☐
		WDC User	☐
		Data.com User Type	None
		Data.com Monthly Addition Limit	Default Limit (300)
		Accessibility Mode (Classic Only)	☐

3. Click **Save & New**.

4. Create the test user for the remote mobile engineering workforce:

- **First Name:** Field | **Last Name:** Technician
- **Email / Username:** Use a unique format (e.g., field_@greengrid.dev).
- **Role:** Assign to the Field Technicians node.
- **User License:** Salesforce
- **Profile:** Standard User

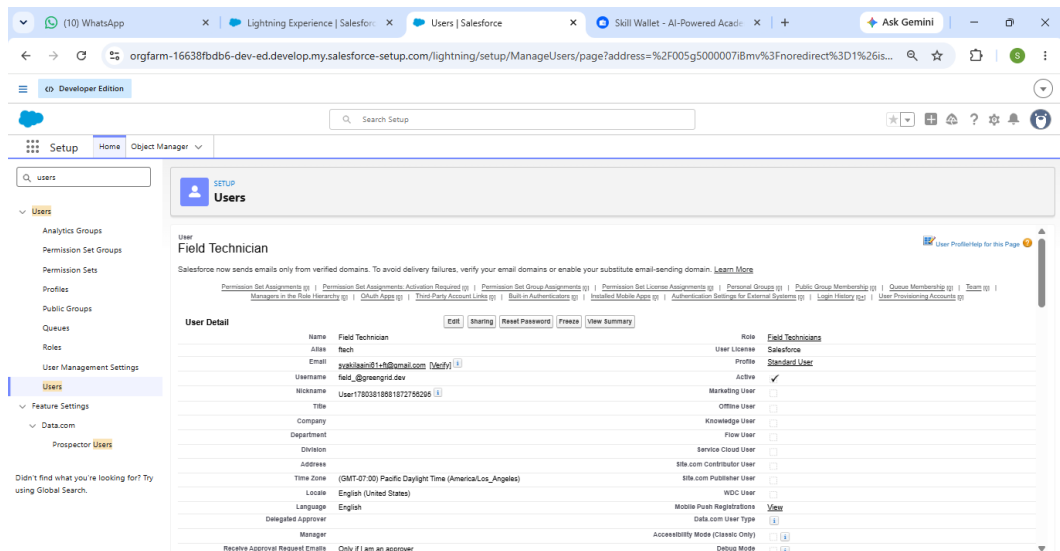
New User

Salesforce now sends emails only from verified domains. To avoid delivery failures, verify your email domains or enable your substitute email-sending domain. [Learn More](#)

User Edit [Cancel] [Save] [Save & New]

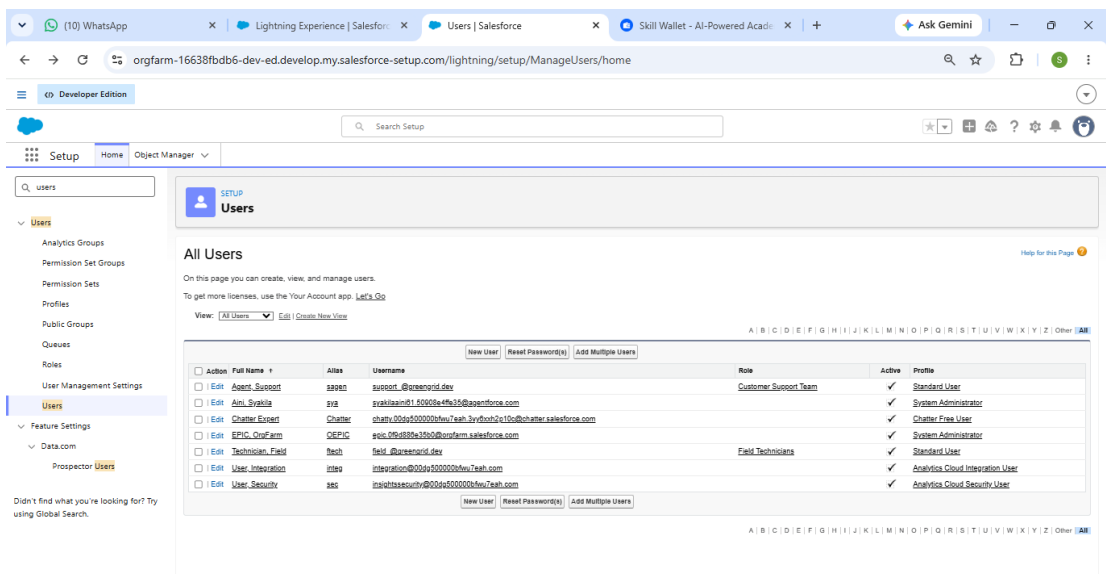
General Information

First Name	Field	Role	Field Technicians
Last Name	Technician	User License	Salesforce
Alias	ftech	Profile	Standard User
Email	syakilaan61+ft@gmail.com	Active	☒
Username	field_@greengrid.dev	Marketing User	☐
Nickname	User178038186818727562	Offline User	☐
Title		Knowledge User	☐
Company		Flow User	☐
Department		Service Cloud User	☐
Division		Site.com Contributor User	☐
		Site.com Publisher User	☐
		WDC User	☐
		Data.com User Type	None
		Data.com Monthly Addition Limit	Default Limit (300)
		Accessibility Mode (Classic Only)	☐



5. Click **Save**.

6. Go back to the **Active Users** list view to check that both the Support Agent and Field Technician entries are active, have the right profiles, and are assigned to the correct roles.



M6. Automation Design with Salesforce Flow

6.1 Flow Design

To effectively address the operational challenges identified in Milestone 2 (B2: Unstructured Dispatch and B3: Post-Resolution Data Latency) while supporting the business goals established in Milestone 1, GreenGrid Utilities adopts a configuration-driven automation strategy using Salesforce Record-Triggered Flows. By eliminating

manual coordination processes and paper-based reporting, the platform enables smooth, real-time integration between customer service operations and field service activities.

6.1.1 Flow 1: Automated Field Dispatch Trigger (Dispatch Engine)

- **Triggering Object:** Case
- **Trigger Source Event:** Record creation or update
- **Trigger Condition Criteria:** Priority EQUALS High AND Status EQUALS New AND Asset_Lookup__c IS NOT NULL TRUE
- **Optimization Goal:** Rapid Field-Service Triage (Actions & Related Records)

Architectural Overview:

When a customer service representative logs an outage request and creates a High-Priority Case associated with a valid Meter ID, this flow automatically removes the need for manual dispatching. It creates a related Field_Visit__c record, connects it to the parent Case, transfers the required customer reference data, and immediately sends a system notification to the field service team.

6.1.2 Flow 2: Real-Time Operational Sync and Case Closure (Sync Engine)

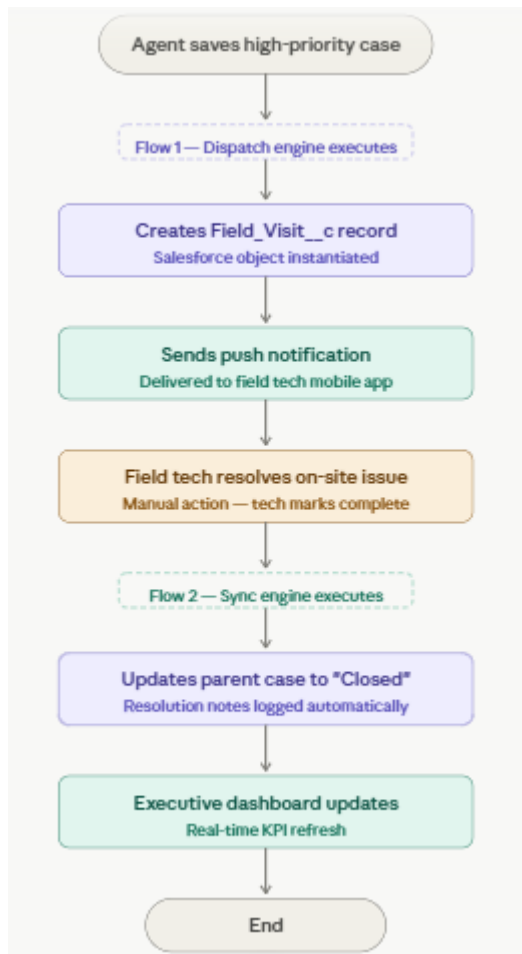
- **Triggering Object:** Field Visit (Field_Visit__c)
- **Trigger Source Event:** Record update
- **Trigger Condition Criteria:** Repair_Status__c EQUALS Resolved AND Repair_Status__c ISCHANGED TRUE
- **Optimization Goal:** Instant Operational Dashboard Visibility (Actions & Related Records)

Architectural Overview:

Once a field technician submits repair details through the Salesforce Mobile App and changes the status to “Resolved,” this automated process transfers the technician’s resolution notes to the related Case record. The flow then updates the Case status to “Closed” and records the completion timestamp, removing reporting delays and providing immediate visibility into operational performance.

6.2 Workflow Documentation

6.2.1 Core Functional Workflows



a. Outage Intake to Automated Technician Dispatch (Flow 1 Workflow)

- 1. User Action:** A Customer Support Agent receives an outage report, performs a Meter ID lookup to associate the Utility_Asset__c record, completes the Case form with Priority = High and Status = New, and saves the record.
- 2. Automated Evaluation:** The Salesforce processing engine validates the transaction and confirms that the Case meets the criteria defined for Flow 1.
- 3. Record Generation:** The flow creates a new Field_Visit__c record, populates Parent_Case__c with the Case ID, assigns Repair_Status__c = Assigned, and references the related customer information.
- 4. Notification Routing:** The platform sends an immediate desktop and mobile notification to the designated field service queue.

b. Mobile On-Site Resolution to Automated Case Closure (Flow 2 Workflow)

- 1. User Action:** After completing the repair work, the Field Technician accesses the record through the Salesforce Mobile App, enters the required

Technician_Notes__c information, updates Repair_Status__c to “Resolved,” and saves the record.

2. **Automated Evaluation:** Salesforce immediately launches Flow 2 when the status change is detected.
3. **Upward Record Update:** The flow updates the related parent Case through Parent_Case__c, sets Status = Closed, and copies Technician_Notes__c into the Case resolution information section.
4. **Metrics Capture:** The flow records Resolution_Timestamp__c on the Field Visit record to support executive reporting and dashboard analytics.

6.3 Error Handling Strategy

To maintain 99.9% platform availability (**NFR1**) and ensure that no critical outage records are lost because of unexpected transaction failures, a structured error-management mechanism is incorporated into every flow design.

6.3.1 Fault Path Routing

Each DML (Data Manipulation Language) operation—including **Create Records and Update Records** elements—is configured with a dedicated **Fault Path Connector**. If issues such as record locking, validation rule failures, or database timeouts occur, Salesforce captures the exception and handles it gracefully without displaying a generic application error to the user.

6.3.2 Exception Notification and Automated Logging

Whenever a fault path is triggered, the platform performs two automated error management procedures:

1. **System Administrator Alerting**

A standard **Send Email** action automatically notifies the IT System Administrator group. The email includes important troubleshooting information such as the Record ID, initiating user profile, execution timestamp, and the system-generated error message (\$Flow.FaultMessage).

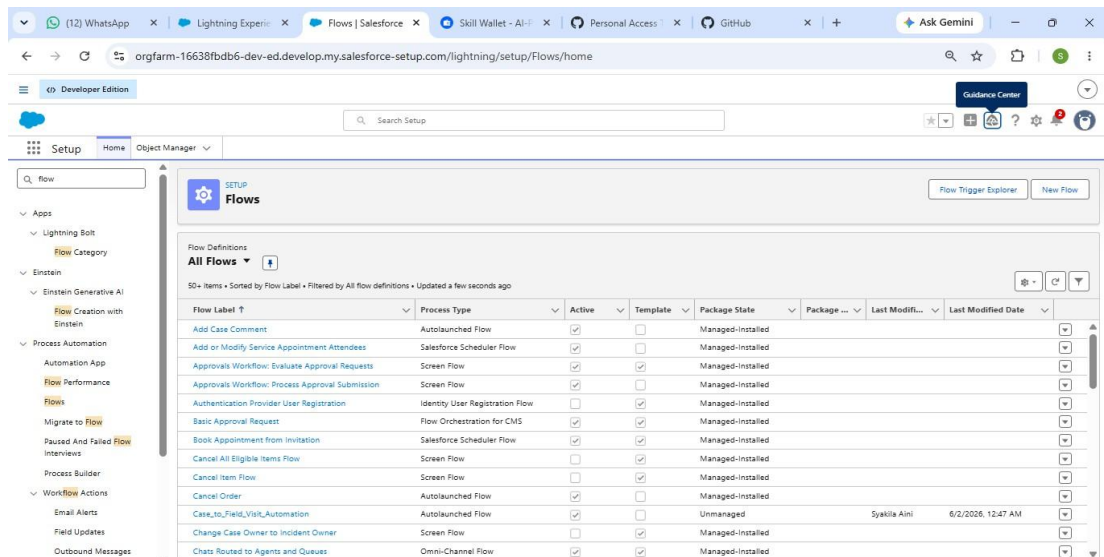
2. **Chatter Fallback Notification**

The flow also publishes a high-priority message to a restricted IT System Administrator Chatter group, ensuring issue visibility and traceability even when email delivery is delayed because of network interruptions.

6.4 Salesforce Implementation and Configuration Steps

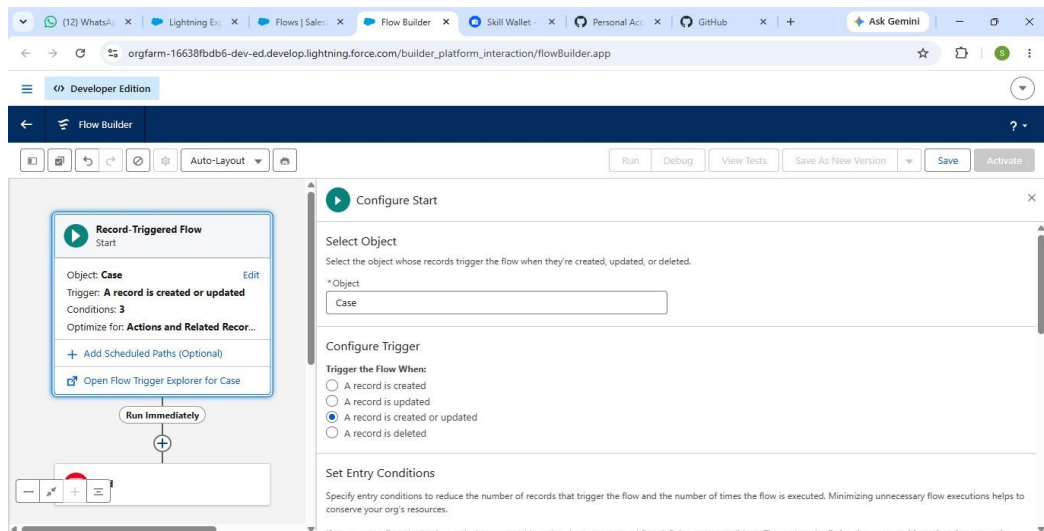
Step 1: Open the Flow Builder Workspace

1. Sign in to your Salesforce Developer environment.
2. Select the gear icon in the upper-right corner and choose **Setup**.
3. Enter **Flows** in the **Quick Find** search box.
4. Select **Flows** from the *Process Automation* section.



Step 2: Create Flow 1 — Automated Field Dispatch Trigger

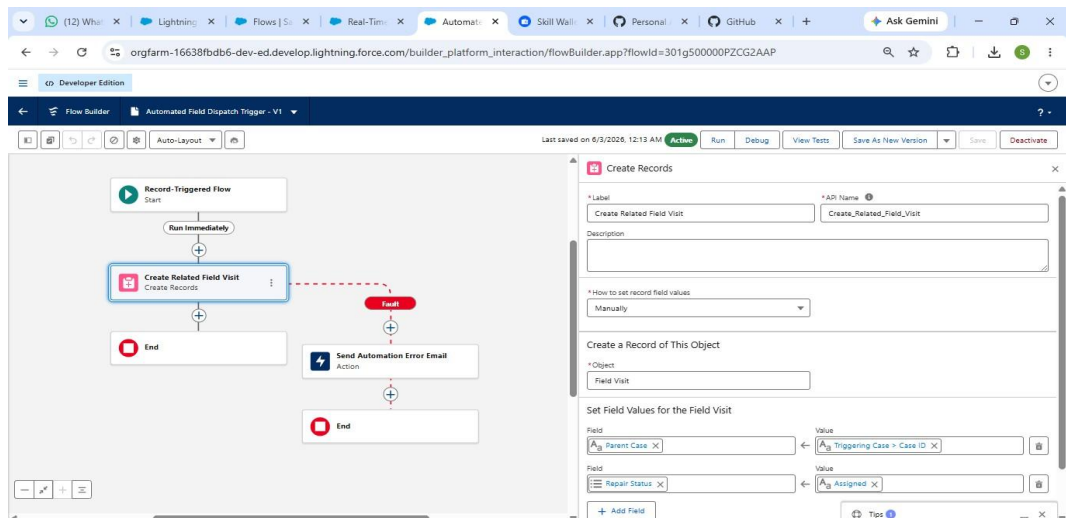
1. On the Flows landing page, click **New Flow**.
2. Choose **Record-Triggered Flow** and click **Create**.
3. Configure the Start settings as follows:
 - **Object:** Case
 - **Trigger the Flow When:** A record is created or updated
 - **Optimize the Flow for:** Actions and Related Records
4. In the **Set Conditions** section, configure:
 - **Condition Requirements:** All Conditions Are Met (AND)
 - **Field 1:** Priority | **Operator:** Equals | **Value:** High
 - **Field 2:** Status | **Operator:** Equals | **Value:** New
 - **Field 3:** Utility_Asset__c | **Operator:** Is Null | **Value:** \$GlobalConstant.False



5. Under **When to Run the Flow for Updated Records**, select:
 - Only when a record is updated to meet the condition requirements.
 Close the configuration window.

Step 3: Add a Create Records Element for Field Visit Creation

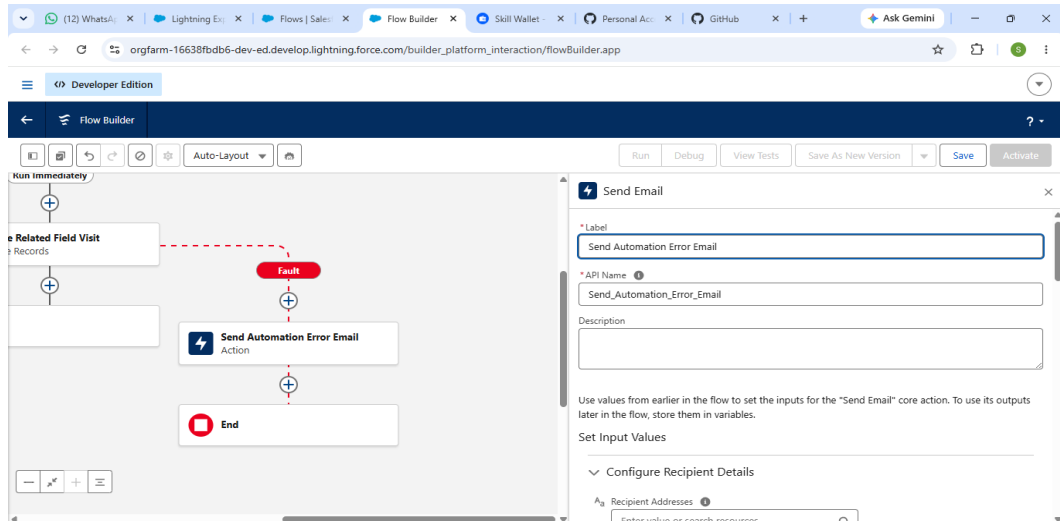
1. Click the plus (+) icon beneath the Start element.
2. Search for and select Create Records.
3. Configure the element:
 - Label: Create Related Field Visit
 - API Name: Create_Related_Field_Visit
 - How Many Records to Create: One
 - How to Set the Record Fields: Use separate resources and literal values
4. Choose:
 - Object: Field_Visit__c
5. Configure field mappings:
 - ****Parent_Case__c**** ← {!\$Record.Id}
 - ****Repair_Status__c**** ← Assigned



6. Save the configuration.

Step 4: Configure Fault Path Error Handling for Flow 1

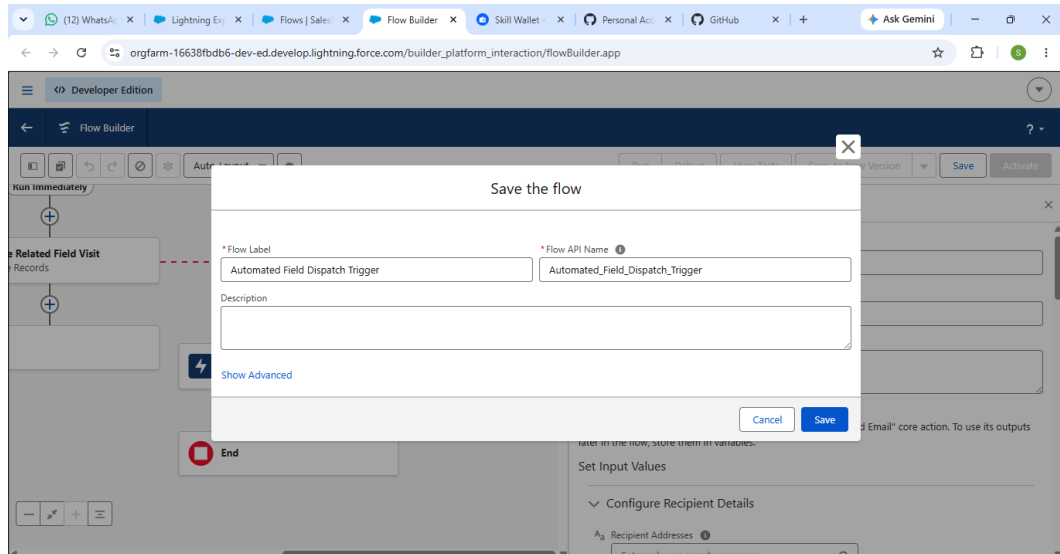
1. Select the **Create Related Field Visit** element.
2. Choose **Add Fault Path**.
3. Click the plus (+) icon on the Fault Path branch.
4. Select **Action**.
5. Search for **Send Email** and select the standard email action.
6. Configure:
 - **Label:** Send Automation Error Email
 - **API Name:** Send_Automation_Error_Email
 - **Body:** The Field Dispatch Flow failed to create a technician dispatch record.
Error Message details: {!\$Flow.FaultMessage}
 - **Subject:** CRITICAL ALERTER: GreenGrid Utilities Salesforce Automation Failure
 - **Email Addresses:** Enter the administrator email address.
 - **Target Workflow Sender:** Set to True



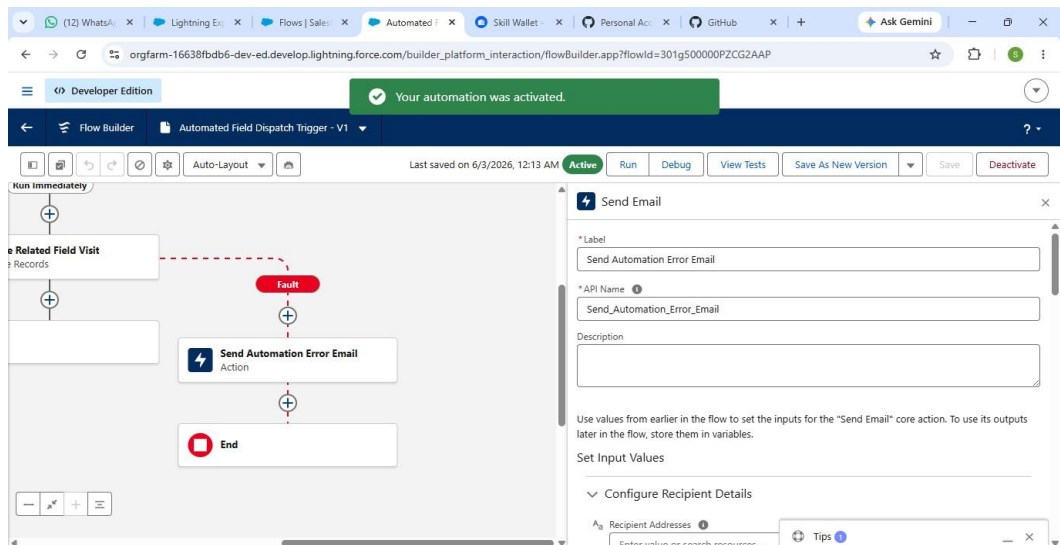
7. Click **Done**.

Step 5: Save and Activate Flow 1

1. Click **Save**.
2. Set the **Flow Label** to Case: Automated Field Dispatch Trigger.
3. Confirm the **Flow API Name** is Case_Automated_Field_Dispatch_Trigger.



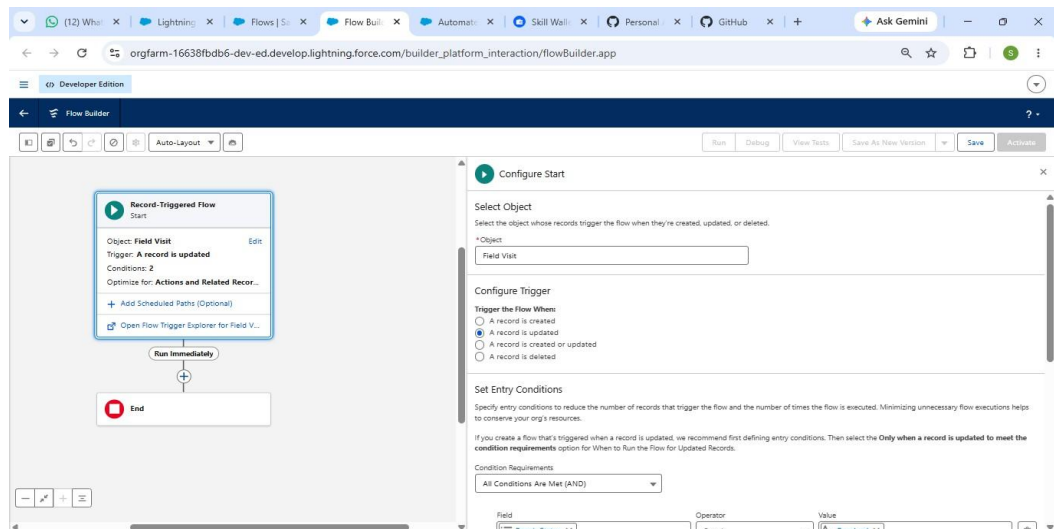
4. Click **Save**.



5. Select **Activate**.
6. Return to Setup.

Step 6: Create Flow 2 — Real-Time Operational Sync and Case Closure

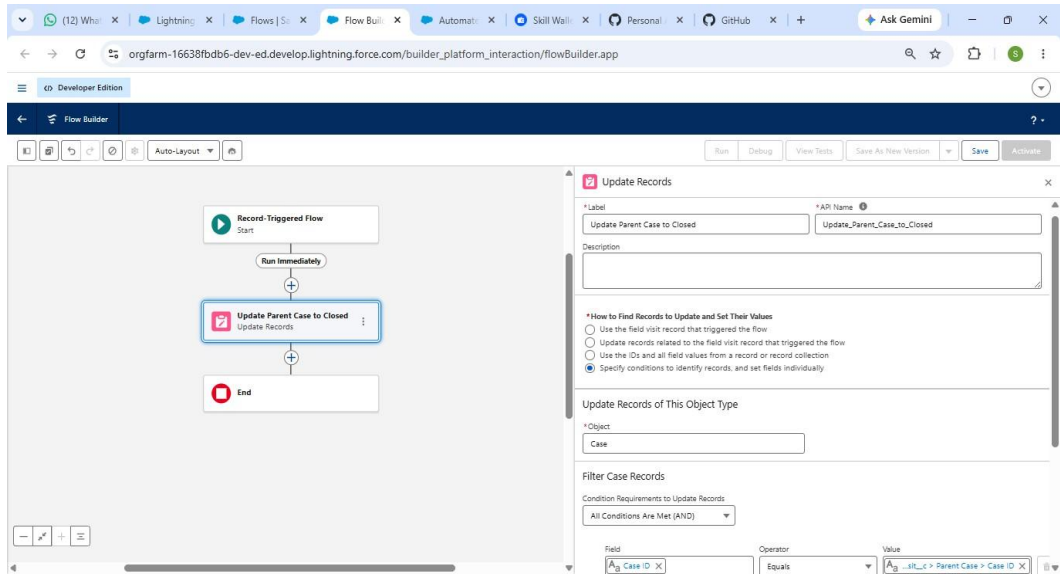
1. From the Flows page, click **New Flow**.
2. Select **Record-Triggered Flow** and click **Create**.
3. figure:
 - **Object:** Field_Visit_c
 - **Trigger the Flow When:** A record is updated
 - **Optimize the Flow for:** Actions and Related Records
4. Set the following conditions:
 - **Condition Requirements:** All Conditions Are Met (AND)
 - **Field 1:** Repair_Status_c | **Operator:** Equals | **Value:** Resolved
 - **Field 2:** Parent_Case_c | **Operator:** Is Null | **Value:** \$GlobalConstant.False
5. Choose:
 - Only when a record is updated to meet the condition requirements.



Close the configuration window.

Step 7: Add an Update Records Element for Case Synchronization

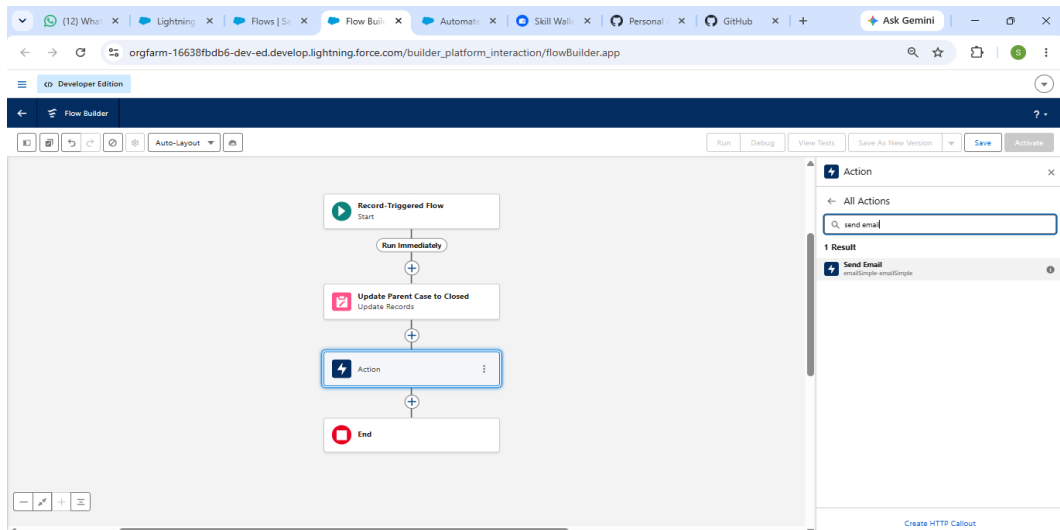
1. Click the plus (+) icon below the Start element.
2. Select **Update Records**.
3. Configure:
 - **Label:** Update Parent Case to Closed
 - **API Name:** Update_Parent_Case_to_Closed
 - **How to Find Records to Update and Set Their Values:** Specify conditions and set fields individually
4. Choose:
 - **Object:** Case
5. Configure the filter:
 - **Field:** Id
 - **Operator:** Equals
 - **Value:** {!\$Record.Parent_Case_r.Id}
6. Configure field updates:
 - **Status** ← Closed
 - **Description** ← {!\$Record.Technician_Notes_c}



7. Save the configuration.

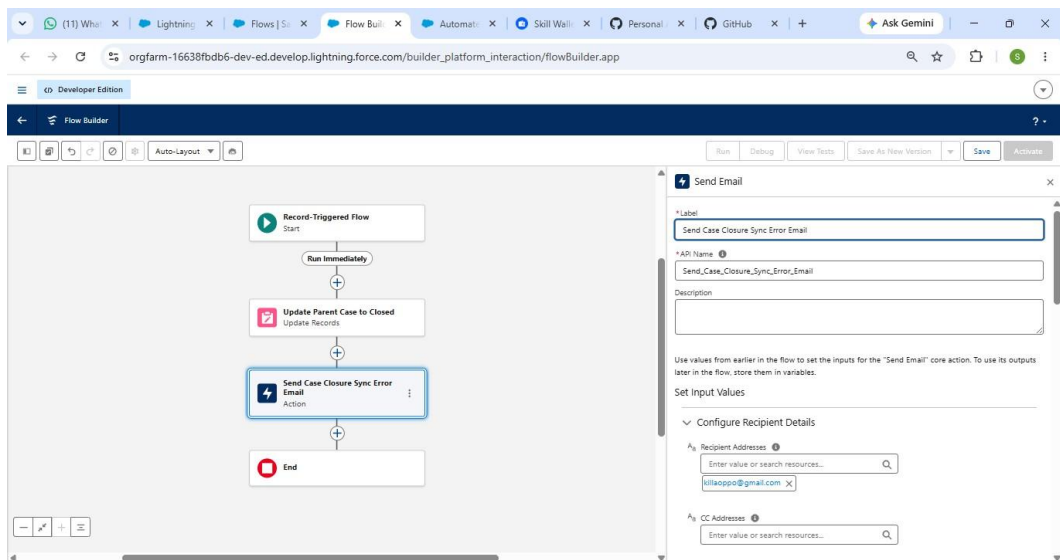
Step 8: Configure Fault Path Error Handling for Flow 2

1. Select **Update Parent Case to Closed**.
2. Choose **Add Fault Path**.
3. Click the plus (+) icon on the fault branch.
4. Select **Action**.
5. Search for and select **Send Email**.



6. Configure:
 - **Label:** Send Case Closure Sync Error Email

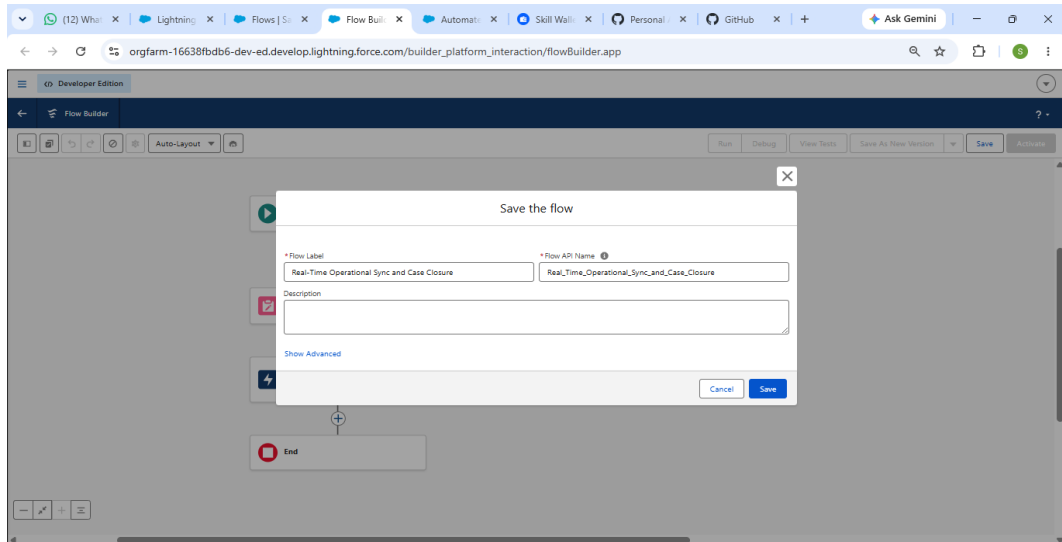
- **API Name:** Send_Case_Closure_Sync_Error_Email
- **Body:** The Operational Sync Flow failed to close the parent Case record. Error Details: {!\$Flow.FaultMessage}
- **Subject:** CRITICAL ALERTER: GreenGrid Utilities Case Closure Failure
- **Email Addresses:** Enter the administrator email address.



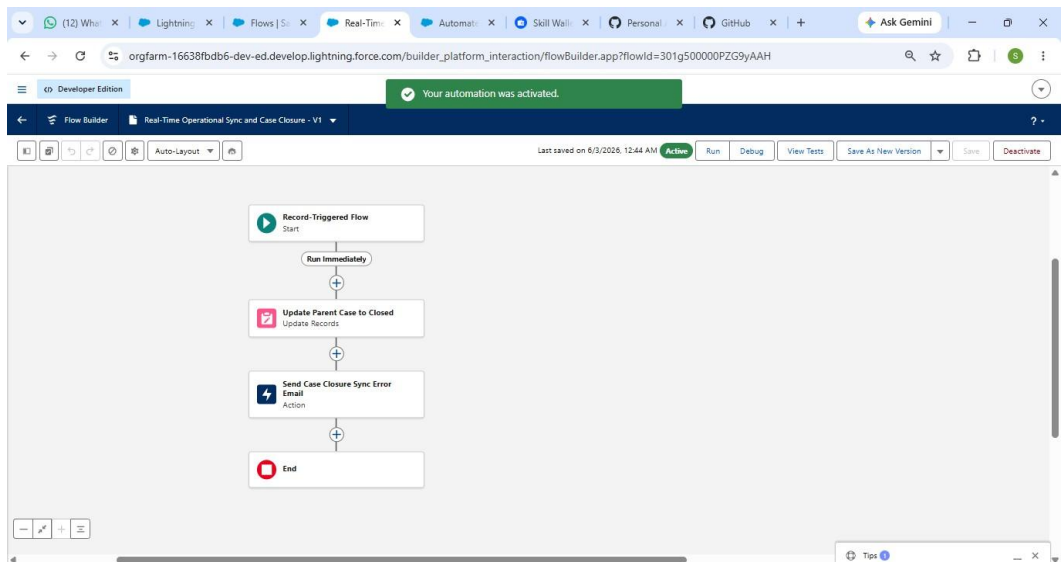
7. Click **Done**.

Step 9: Save and Activate Flow 2

1. Click **Save**.
2. Set the **Flow Label** to *Field Visit: Real-Time Operational Sync and Case Closure*.
3. Verify the **Flow API Name** is *Field_Visit_Real_Time_Operational_Sync_and_Closure*.



4. Click **Save**.



5. Select **Activate**.

Step 10: Perform End-to-End Functional Testing

1. Open the **Cases** tab from the App Launcher. Create a test Case with Priority = High and Status = New, associate it with an existing Account and Utility_Asset__c record, and save it.
2. Confirm that a related **Field Visit** record is automatically generated under the Case's related list, indicating that **Flow 1** executed successfully.
3. Open the generated **Field Visit** record, click **Edit**, enter information in ****Technician_Notes__c**** (for example, "Grid line 4 repair completed safely."), change ****Repair_Status__c**** to **Resolved**, and save

the record.

4. Return to the parent **Case** record and verify that the status has automatically changed to **Closed** and that the technician notes appear in the Case details, confirming successful execution of **Flow 2**.

Step 11: Testing and Verification (Complete Guide)

1. Creating a New Utility Asset

- a. Open the **GreenGrid Utilities** application, navigate to the **Utility Assets** tab, and click the **New** button.
- b. Populate the information in the *New Utility Asset* window with the following details:
 - **Asset Name:** (Leave blank for the system to automatically generate a serialized sequence, e.g., MET-0002 or MET-0003).
 - **Meter ID:** Enter the required Meter ID (e.g., MTR-2026-002).
 - **Asset Status:** Active.
 - **Installation Address:** Enter the appropriate installation address.
 - **Customer Account:** Search and select the associated customer account (e.g., Residential Account - Lucky Pradana).
- c. Click **Save**. Once saved, verify that the *Asset Name* field automatically generates a unique serialized sequence.

The screenshot shows a web browser window with the URL `orgfarm-16638fbd6-dev-ed.develop.lightning.force.com/lightning/o/Account/list?filterName=__Recent`. The application interface includes a top navigation bar with tabs for 'GreenGrid Utilities', 'Accounts', 'Contacts', 'Cases', 'Utility Assets', and 'Field Visits'. The 'Accounts' tab is active, displaying a 'Recently Viewed' section with 2 items. The table below lists the accounts:

	Account Name	Account Site	Phone	Account Owner Alias
1	Residential Account - Lucky Pradana		085310167098	sya
2	Residential Account - Anton Wilbowo		083165441230	sya

Information

Asset Name

Meter ID: MTR-285KB-002

Asset Status: Active

Installation Address: Jl. Gatot Subroto No. 45

Customer Account: Residential Account - Lucky Pratama

Owner: Syakila Aini

Buttons: Cancel, Save & New, Save

Click Save.

Utility Asset: MET-0002

Related Details

Asset Name: MET-0002

Meter ID: MTR-285KB-002

Asset Status: Active

Installation Address: Jl. Gatot Subroto No. 45

Customer Account: Residential Account - Lucky Pratama

Owner: Syakila Aini

Created By: Syakila Aini, 6/9/2026, 1:04 AM

Last Modified By: Syakila Aini, 6/9/2026, 1:04 AM

Buttons: New Contact, Edit, New Opportunity

2. Simulating a New Customer Case

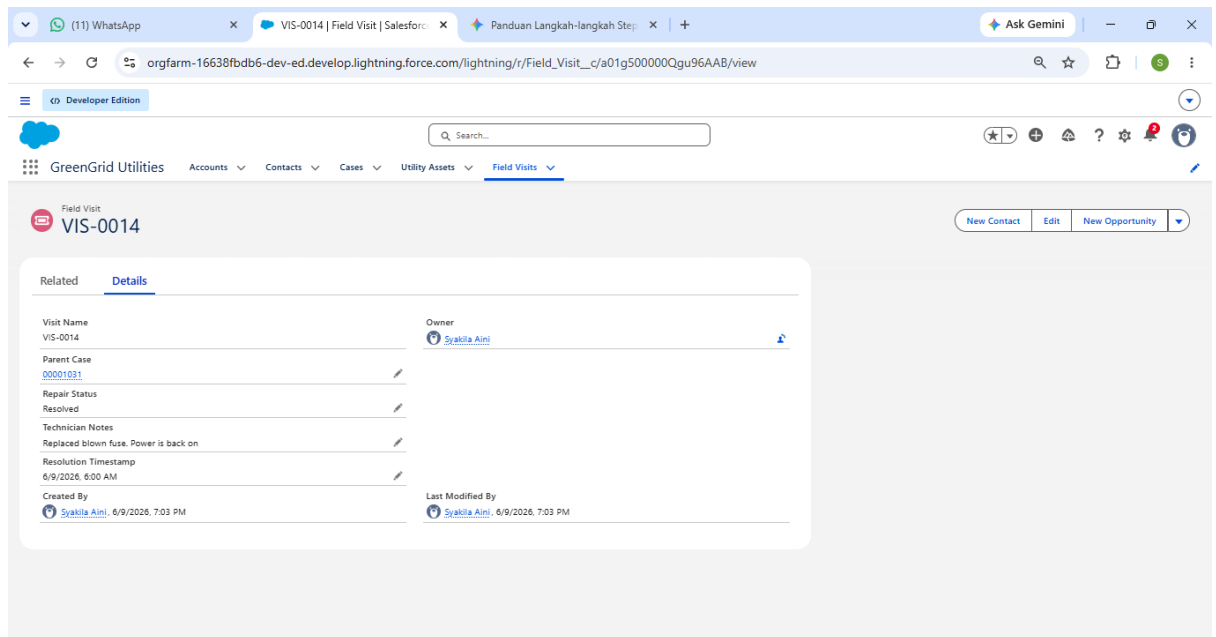
- Navigate to the **Cases** navigation tab on the main menu bar and click the **New** button located at the top right to simulate a live customer grid infrastructure disruption log.
- Populate the *New Case* form on the screen with the following data:
 - Account Name:** Residential Account - Lucky Pradana
 - Contact Name:** Lucky Pradana
 - Utility Asset:** Select the newly created asset associated with this account (e.g., MET-0002).

- **Status:** New
- **Priority:** High
- **Case Origin:** Select Email (or Phone depending on the live customer call channel being simulated).
- **Case Type (Type):** Outage

c. Click **Save**.

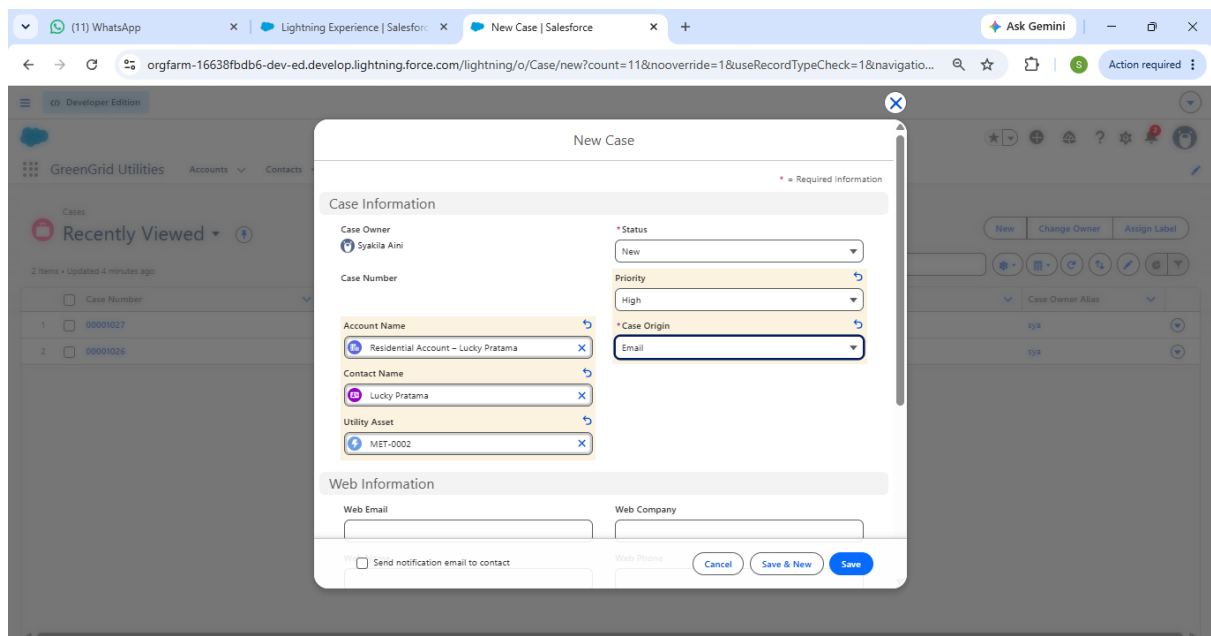
3. System Automation Validation (Field Visits Generation)

- Once the Case record updates on the screen, scroll down to the **Field Visits** related list block on the page layout, or check the global **Field Visits** navigation tab.
- **Validation:** Verify that a brand-new child record (e.g., VIS-0001 or VIS-0002) with an operational repair status of **Assigned** has been generated instantly by the background Salesforce Flow engine without any manual user interaction, confirming absolute reference integrity across the data channel.

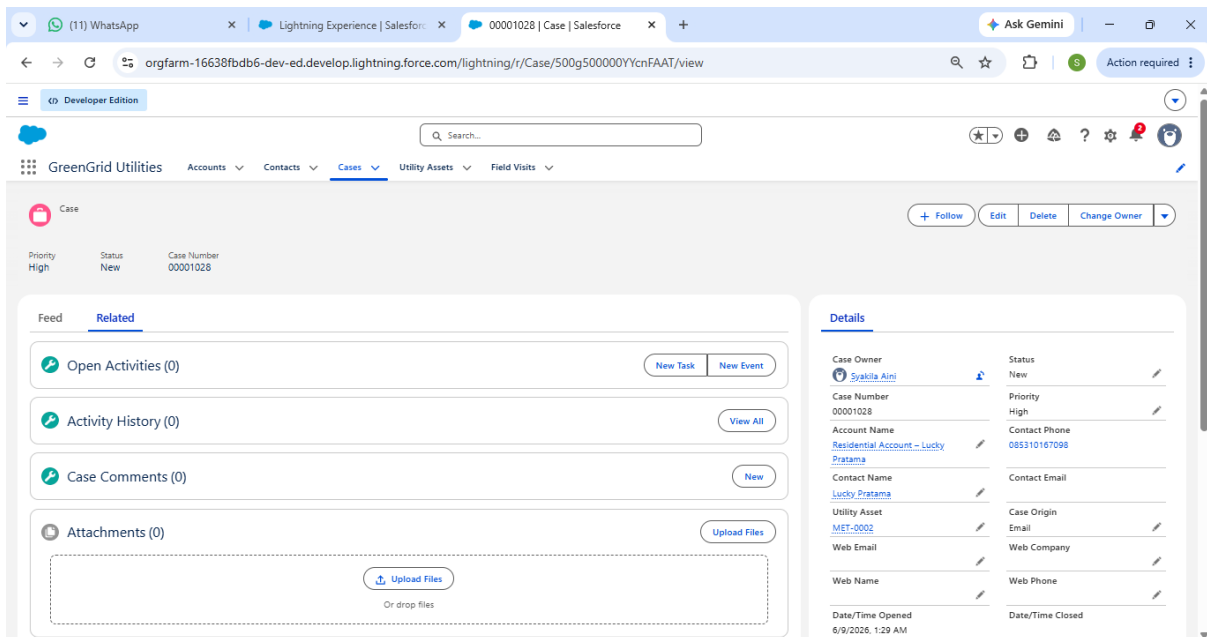


4. Final Case Resolution and Closure Verification

- After the field technician completes the required operational work, return to the parent **Case** record.
- Verify that the case status has automatically changed to **Closed** and that the technician notes appear correctly within the Case details, confirming the successful and complete execution of **Flow 2**.



Click Save.



M7. Reporting and Analytics Design

7.1 Report Definition

To fulfill the primary business goals of enabling real-time management visibility (Milestone 1) and evaluating operational performance improvements over the legacy spreadsheet system (Milestone 2), GreenGrid Utilities makes use of Salesforce's built-in reporting engine.

Reports are organized through specific report types that span both standard and custom object boundaries, ensuring accurate data aggregation without redundancy.

7.1.1 Core Operational Report Specifications

a. Report 1: Average Handling Time (AHT) and Case Triage Performance

- **Report Type:** Cases (Standard Report Type)
- **Format:** Summary Report (Grouped by Owner and Priority)
- **Core Metrics:** Average Age (Hours/Minutes), Total Case Count, and Case Status Breakdown.
- **Business Justification:** Directly monitors BR1 (targeting a 25% reduction in AHT). It reveals how efficiently agents log, process, and transfer incoming utility outage calls within the Salesforce Service Console, compared to the 5-minute delay caused by the legacy spreadsheet system (B1).

b. Report 2: Field Visit Resolution Efficiency

- **Report Type:** Cases with Field Visits (Custom Report Type built on a Case parent and Field_Visit__c child relationship)
- **Format:** Matrix Report (Rows grouped by Field_Visit__c.Owner / Field Technician; Columns grouped by Field_Visit__c.Repair_Status__c)
- **Core Metrics:** Record Count (Total Dispatches), Field Visit Success Rate (Ratio of Resolved vs. Failed), and Average Turnaround Time.
- **Business Justification:** Evaluates field operational performance (BR3 — targeting a 15% improvement in first-visit fix rate) and tracks the transition away from informal, untracked messaging platforms (B2).

c. Report 3: High-Priority Grid Outage Trends

- **Report Type:** Cases with Utility Assets (Custom Report Type built on a Case parent and Utility_Asset__c child relationship)
- **Format:** Summary Report (Grouped by Utility_Asset__c.Asset_Status__c and Asset Installation Address/Region)
- **Core Metrics:** Total High-Priority Cases, Outage Duration Tracking, Asset Failure Frequency.
- **Business Justification:** Gives Executive Leadership instant visibility into which critical grid hardware components are experiencing repeated service disruptions.

7.2 Dashboard Design

Dashboards serve as the central command interface for GreenGrid Utilities' leadership tier (Apex Tier in the Role Hierarchy). They transform rows of raw report data into real-time, easily scannable visual performance components.

7.2.1 Executive Operations Dashboard Blueprint

Component 1 (Top Left) – Open Crisis Metric

A large single-number metric component displaying the active count of High-Priority Cases where Status = New or Working. It refreshes instantly whenever Flow 1 (Dispatch Engine) runs.

Component 2 (Top Center) – Service Console Speed Gauge

A visual gauge tracking Average Case Age. The green zone represents optimal handling speed below 2 minutes, directly demonstrating the resolution of spreadsheet bottleneck B1.

Component 3 (Top Right) – First-Visit Fix Ratio Donut

A donut chart displaying the breakdown of Field_Visit__c.Repair_Status__c values (Assigned, In Progress, Resolved, Failed). This measures the elimination of paper log losses (B3).

Component 4 (Bottom Main) – Grid Outage Distribution Bar Chart

A vertical bar chart tracking Case counts grouped by Asset Installation Address. This supports executive planning for physical infrastructure improvements.

7.3 Data Visualization Strategy

To maintain the performance and configuration standards outlined in NFR3 (Dashboards must load within a 5-second window), GreenGrid Utilities enforces strict data visibility rules and access controls for dashboards.

7.3.1 Role Hierarchy Integration & Dashboard View Context

- Dashboards are set to run under a designated administrative user context ("View Dashboard As: Specified User" assigned to an Executive Manager).
- This setup allows the system to pre-cache complex multi-object roll-ups across the entire organization. It avoids the performance delays caused by real-time role-based sharing recalculations during live presentations, in accordance with NFR3 load-time requirements.

7.3.2 Foldering and Security Governance

- Reports and dashboards are stored within a restricted folder environment named **Executive Operational Analytics**.
- **Access Control:** The folder is granted **Viewer** access to the Customer Support Team and Field Technicians role groups to reduce workspace clutter, and **Manage/Editor** access to Executive Management and IT/System Administrators.

7.4 Salesforce Implementation and Configuration Steps

Follow these step-by-step instructions to define custom report structures, build analytical components, assemble the executive dashboard layout, and secure the operational folders.

Step 1: Create Custom Report Types for Multi-Object Relationships

Before building reports, custom frameworks must be deployed to connect custom entities (Utility_Asset__c and Field_Visit__c) with standard objects.

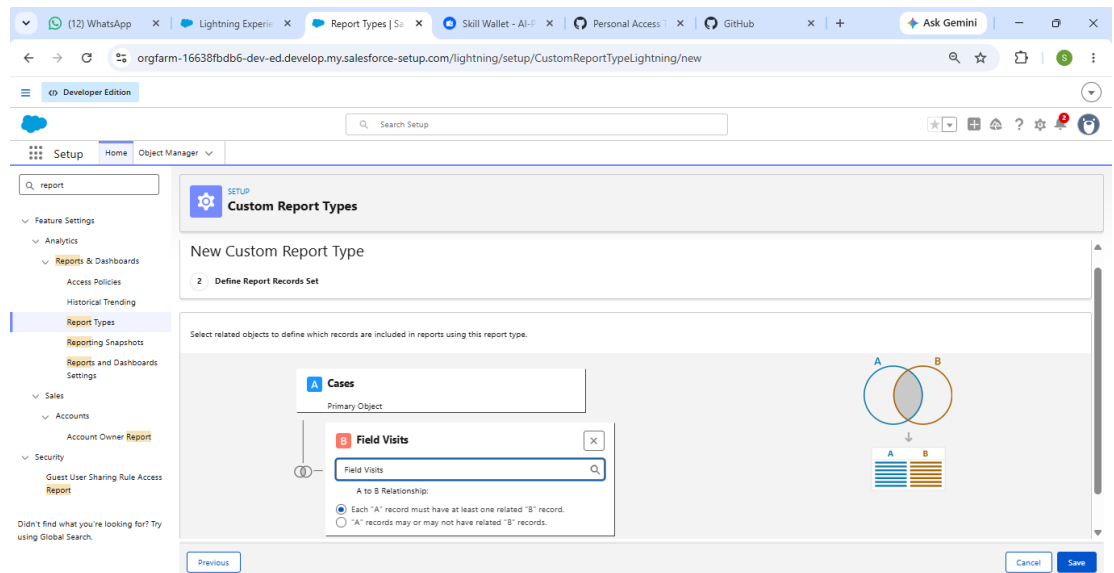
- Navigate to **Setup** by clicking the gear icon in the top-right corner.
- In the Quick Find box on the left, type **Report Types** and select **Custom Report Types**.
- If an introductory screen appears, click **Continue**.
- Click **New Custom Report Type**.
- Fill in the Report Type settings as follows:
 - **Primary Object:** Select Cases.
 - **Report Type Label:** Type Cases with Field Visits.
 - **Report Type Name:** Auto-populates as Cases_with_Field_Visits.
 - **Description:** Type *Operational report linking customer service tickets to field crew dispatch actions*.
 - **Store in Category:** Select Customer Support Cases.
 - **Deployment Status:** Select Deployed.

The screenshot shows the Salesforce Setup interface for creating a new Custom Report Type. The browser address bar indicates the URL: `orgfarm-16638fbd6-dev-ed.develop.my.salesforce-setup.com/lightning/setup/CustomReportTypeLightning/new`. The left sidebar shows the navigation menu with 'Report Types' selected under 'Reports & Dashboards'. The main content area is titled 'Custom Report Types' and contains the following fields:

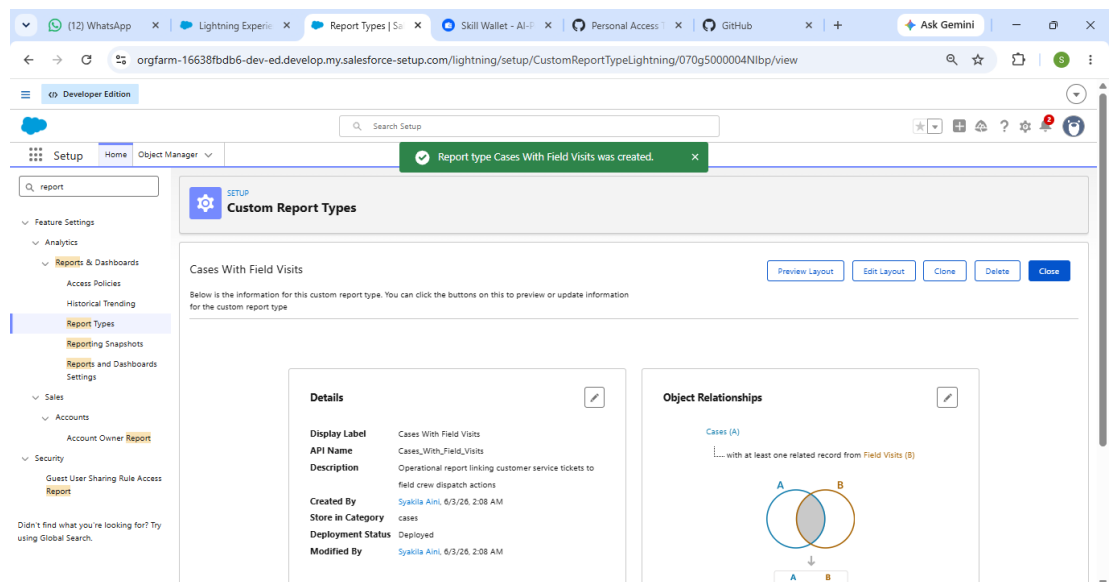
- Primary Object:** A dropdown menu with 'Cases' selected.
- Details:**
 - *Display Label:** A text field containing 'Cases With Field Visits'.
 - *API Name:** A text field containing 'Cases_With_Field_Visits'.
 - *Description:** A text area containing 'Operational report linking customer service tickets to field crew dispatch actions'.
 - *Store in Category:** A dropdown menu with 'Customer Support Reports' selected.

At the bottom right, there are 'Cancel' and 'Next' buttons.

- Click **Next**.
- On the Define Object Relationships canvas, click the empty box below A (Cases) to add a child object.
- Select **Field Visits (Field_Visit__c)** from the dropdown. Leave the default setting: *Each "A" record must have at least one related "B" record.*



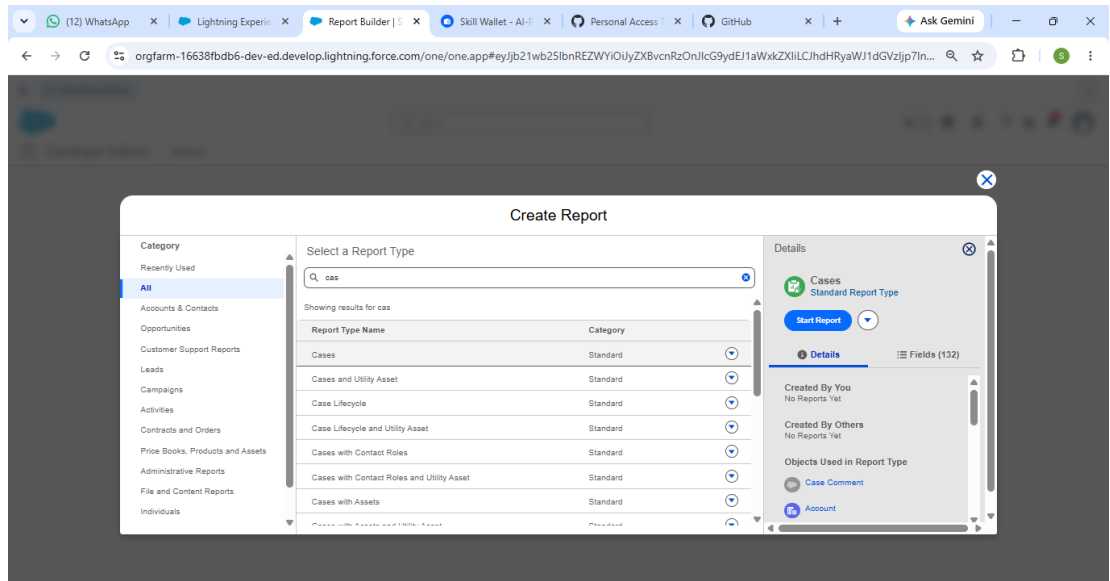
- Click **Save**.



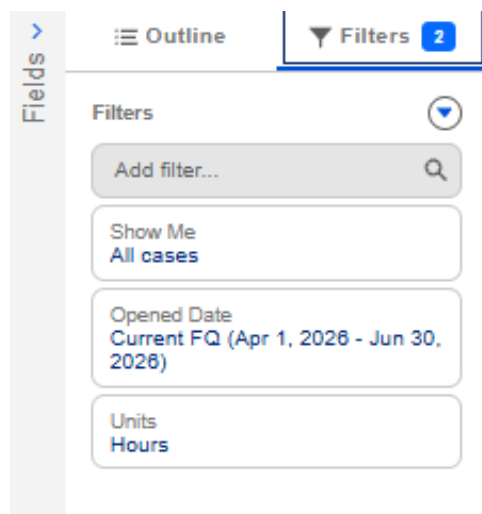
- (Optional Repeat for Assets) Click **New Custom Report Type** again. Create a type with Cases as Primary, link Utility Assets (Utility_Asset__c) as object B, label it **Cases with Utility Assets**, set it to Deployed, and click **Save**.

Step 2: Build Report 1 — Average Handling Time (AHT) Analytics

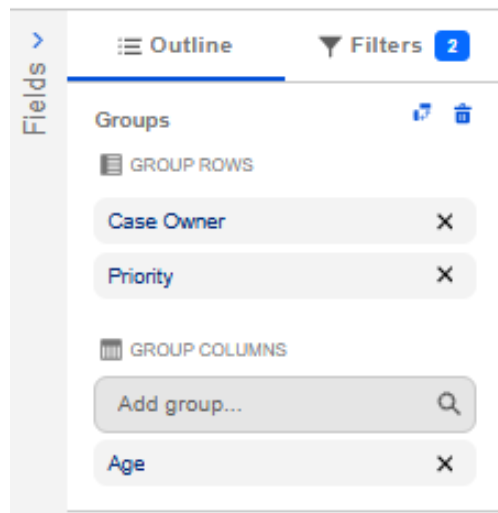
- Click the **App Launcher** icon and search for **Reports**.
- Click **Reports**, then click **New Report**.
- Search for and select the standard **Cases** report type.
- Click **Start Report**.



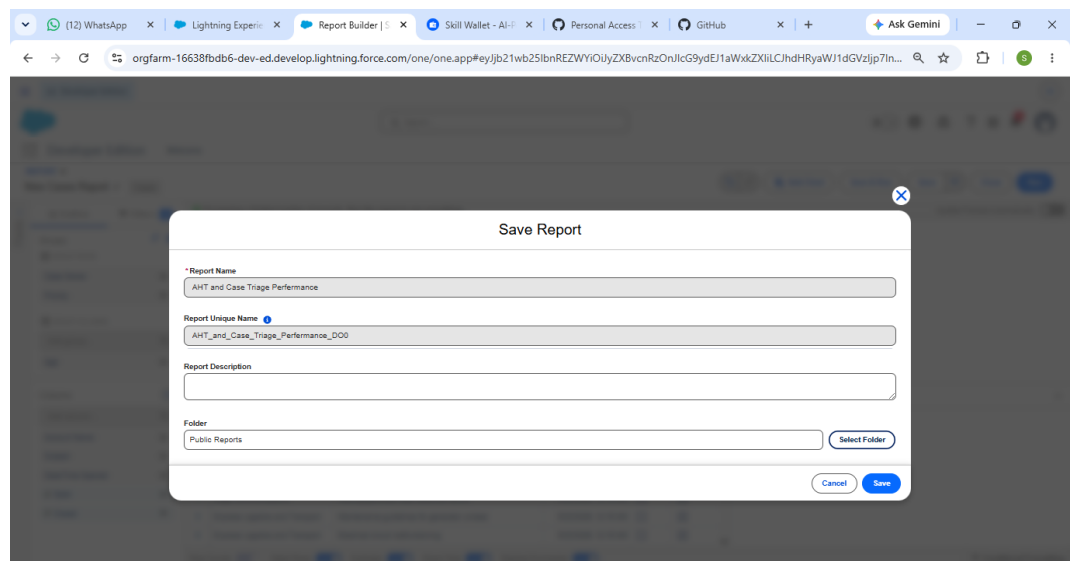
- Enable **Update Preview Automatically**.
- Under **Filters**:
 - Change **Show Me** from My Cases to **All Cases**.
 - Set **Opened Date** to Current FQ (or All Time for testing).



- Under **Outline:**
 - Group Rows by **Case Owner**.
 - Add a second grouping by **Priority**.



- Add the **Age** column and remove unnecessary fields.
- Summarize the Age field using **Average**.
- Click **Save & Run**.
- Save as:
 - **Report Name:** AHT and Case Triage Performance
 - **Unique Name:** AHT_and_Case_Triage_Performance



Report: Cases AHT and Case Triage Performance

Total Records: 26

Case Owner	Priority	Age	-9.972	278	Total
Orgfarm EPIC	High	Record Count	6	0	6
	Medium	Record Count	13	0	13
	Low	Record Count	4	3	7
	Subtotal	Record Count	23	3	26
Total		Record Count	23	3	26

Details (26 Rows) Click an intersection in the table above to filter details.

Account Name	Subject	Date/Time Opened	Open	Closed
1 Edge Communications	Starting generator after electrical failure	5/22/2026, 12:15 AM	☐	☒
2 Grand Hotels & Resorts Ltd	Delay in installation; spare parts unavailable	5/22/2026, 12:15 AM	☐	☒
3 Burlington Textiles Corp of America	Structural failure of generator base	5/22/2026, 12:15 AM	☐	☒
4 United Oil & Gas Corp.	Generator GC3000 platform structure is weakening	5/22/2026, 12:15 AM	☐	☒
5 United Oil & Gas Corp.	Performance inadequate for second consecutive week	5/22/2026, 12:15 AM	☐	☒

Row Counts: ☐ Detail Rows: ☒ Subtotals: ☒ Grand Total: ☒ Stacked Summaries: ☒

Step 3: Build Report 2 — Field Visit Resolution Efficiency

- Click **New Report**.
- Select **Cases with Field Visits**.

Create Report

Category: Recently Used

Select a Report Type

Showing results for cases with file

Report Type Name	Category
Cases With Field Visits	Custom

Details

Cases With Field Visits
Custom Report Type

Start Report

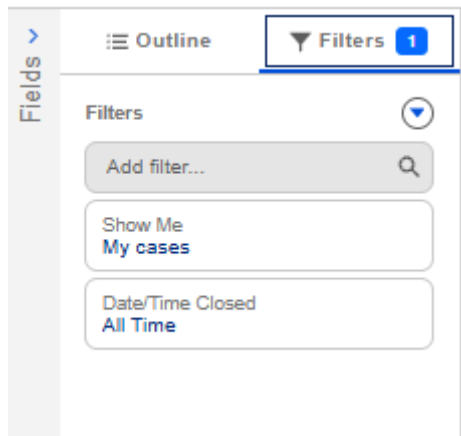
Description
Operational report linking customer service tickets to field crew dispatch actions

Created By You
No Reports Yet

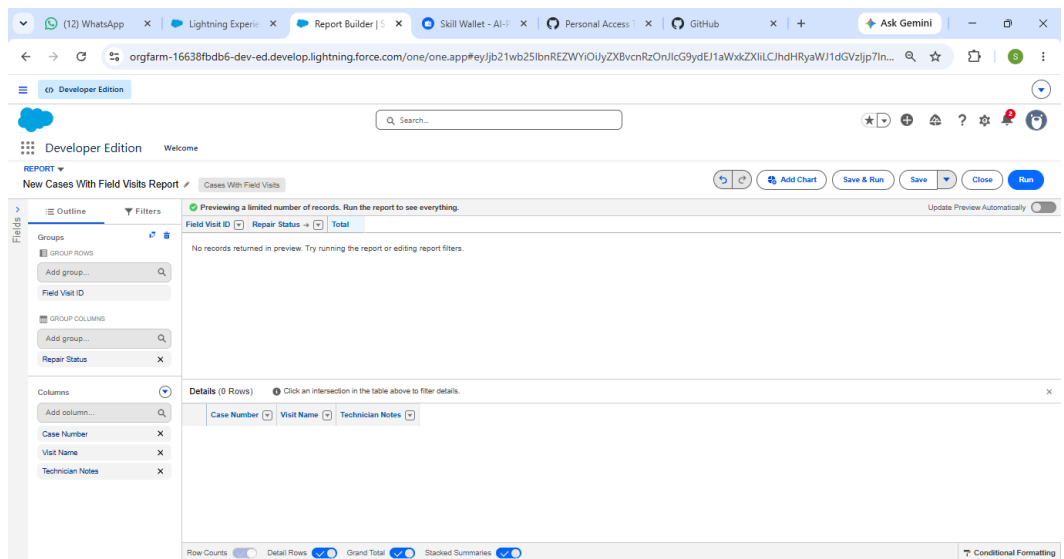
Created By Others
No Reports Yet

Objects Used in Report Type

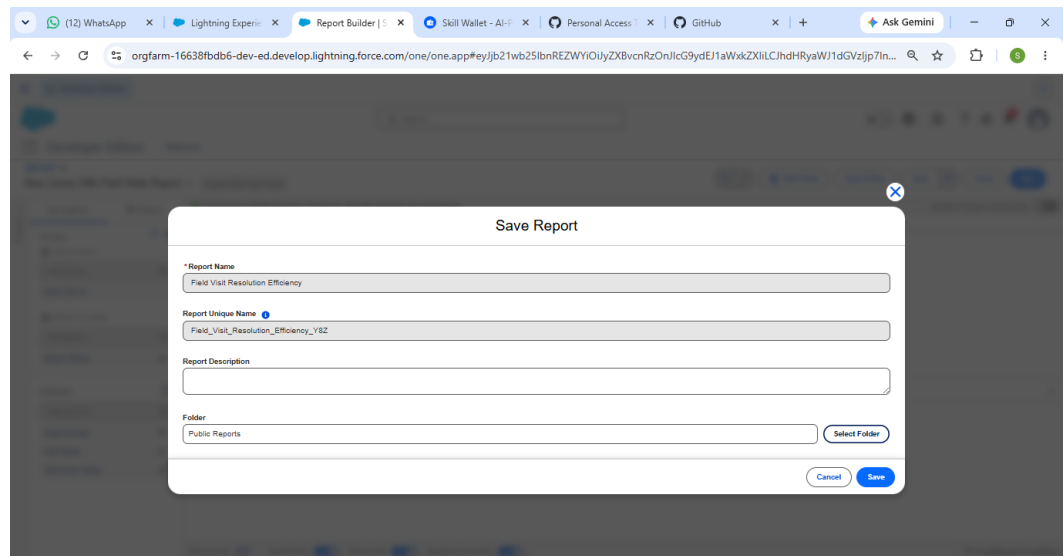
- Click **Start Report**.
- Under **Filters**:
 - **Show Me: All Cases**
 - **Opened Date: All Time**



- Under **Outline:**
 - Group Rows by **Field Visit Owner**
 - Group Columns by **Repair Status**
 - Add: **Visit Number** and **Technician Notes**

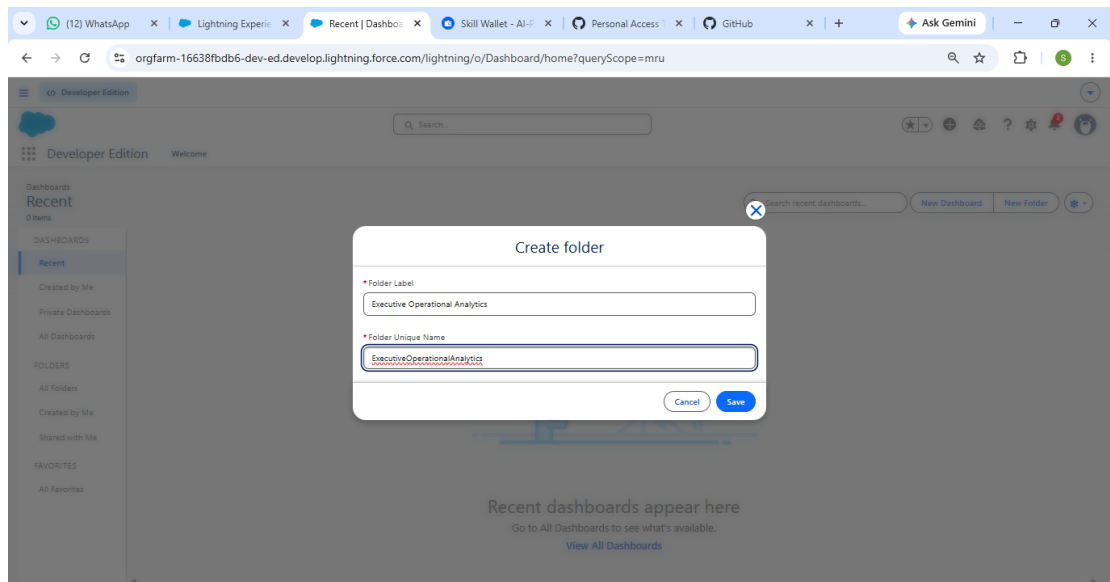


- Click **Save & Run**.
- Save as:
 - **Report Name:** Field Visit Resolution Efficiency
 - **Unique Name:** Field_Visit_Resolution_Efficiency



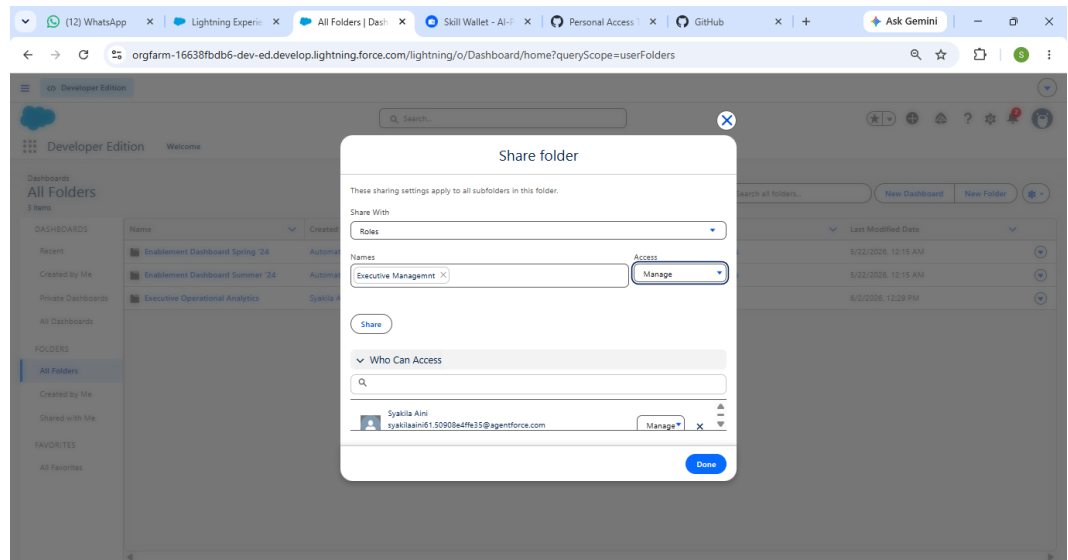
Step 4: Provision Secured Analytics Folders

- Open the **Dashboards** tab.
- Click **New Dashboard Folder**.
- Name the folder: **Executive Operational Analytics**.

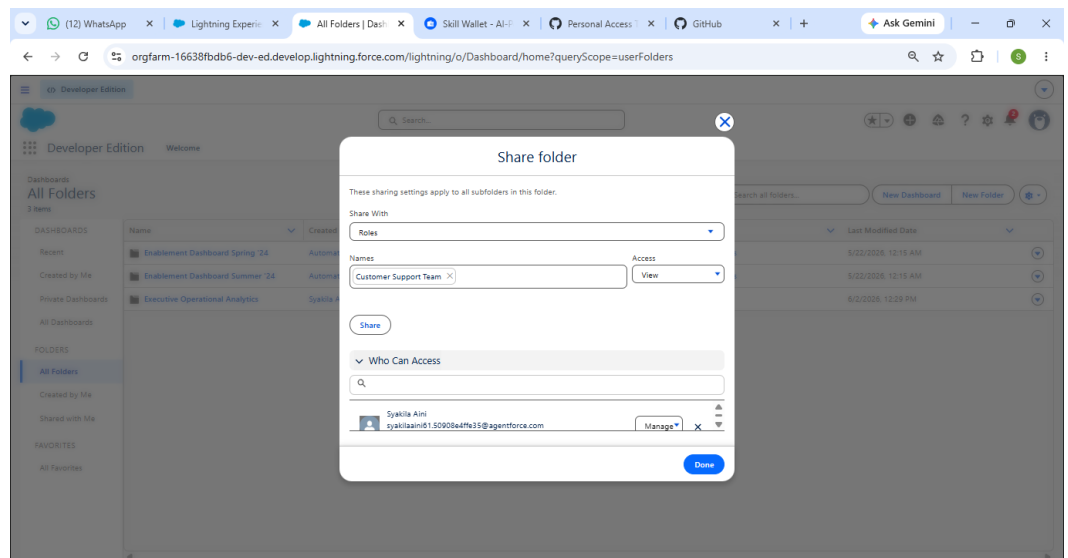


- Click **Save**.
- Open **Share** settings for the folder.
- Configure:

- **Executive Management** → **Manage**



- **Customer Support Team** → **Viewer**

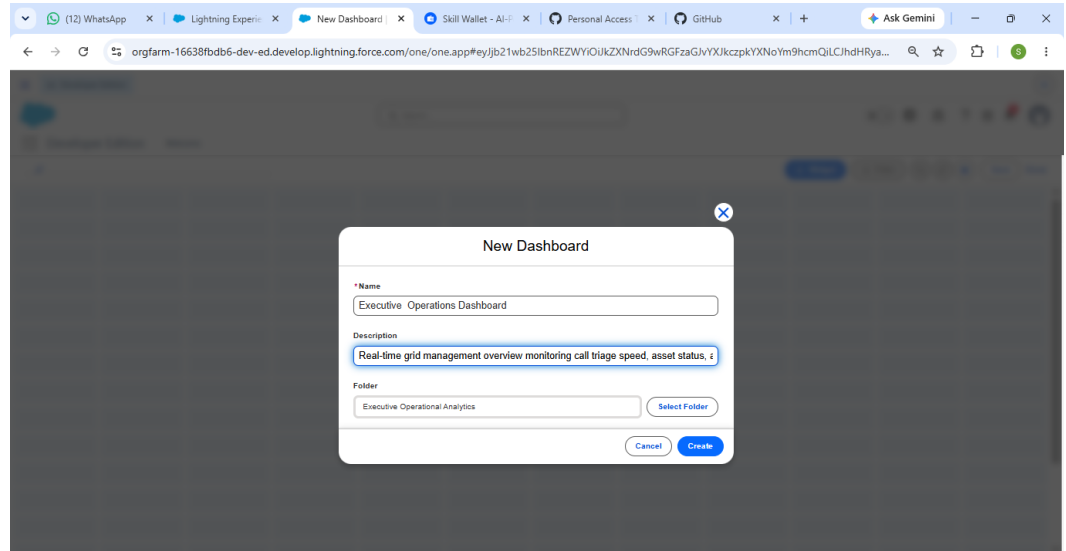


- Click **Done**.

Step 5: Assemble the Executive Operations Dashboard

- Open **Dashboards**.
- Click **New Dashboard**.
- Configure:
 - **Name:** Executive Operations Dashboard

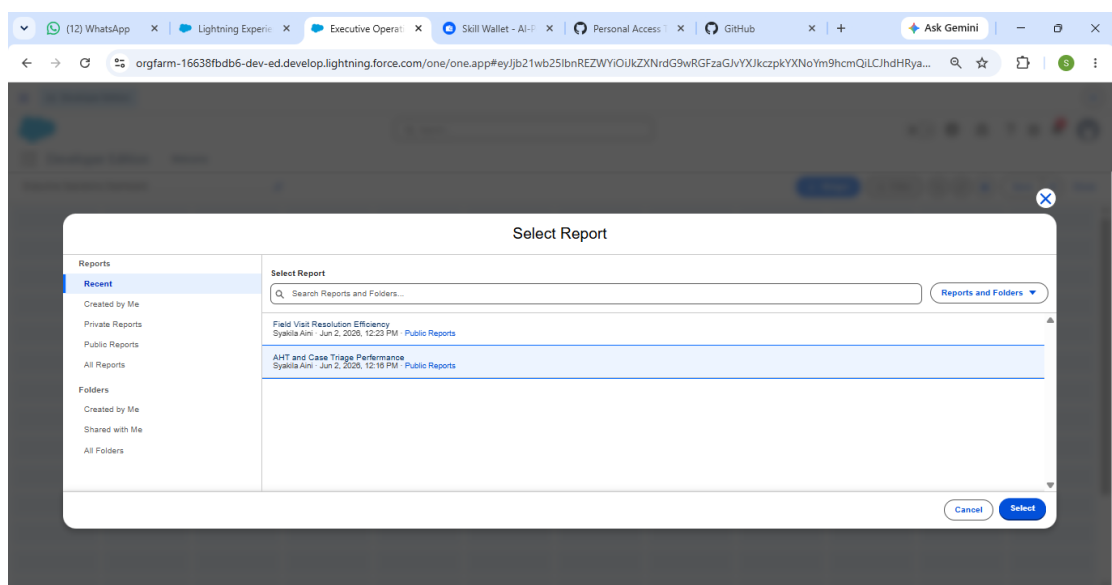
- **Description:** Real-time grid management overview monitoring call triage speed, asset status, and mobile technician field repair rates.
- **Folder:** Executive Operational Analytics



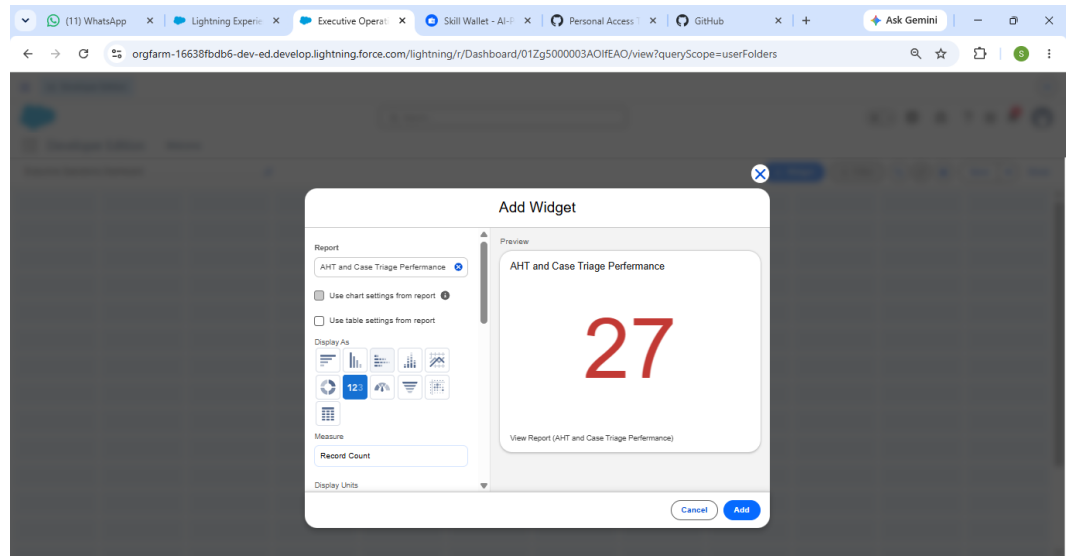
- Click **Create**.

Step 6: Add Component 1 — Live Open Crisis Counter

- Click **+ Component**.
- Select report: **AHT and Case Triage Performance**.



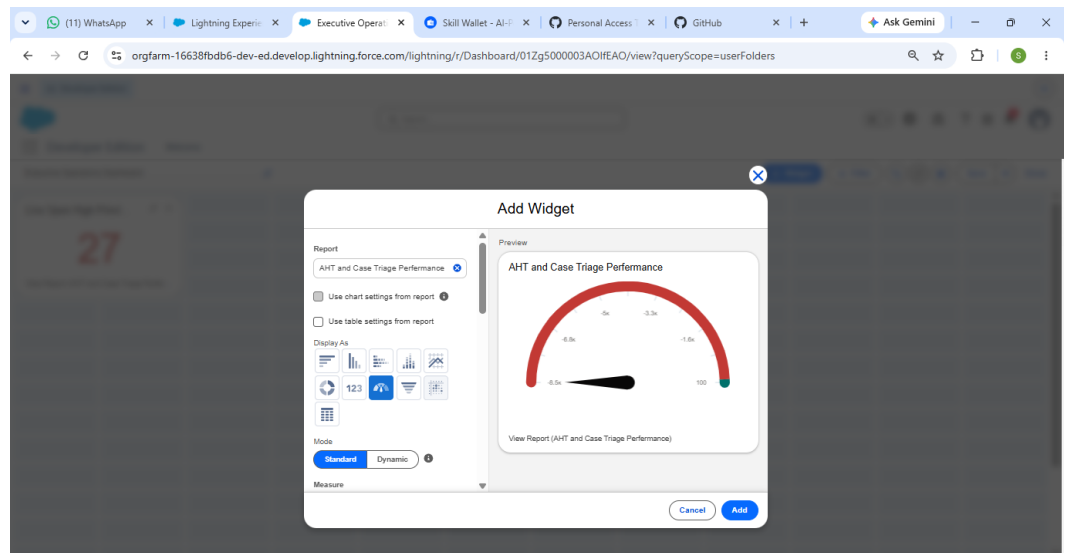
- **Configure:**
 - **Component Type:** Metric
 - **Measure:** Record Count
 - **Title:** Live Open High-Priority Cases



- Click **Add** and position it in the upper-left section.

Step 7: Add Component 2 — Service Console Speed Gauge

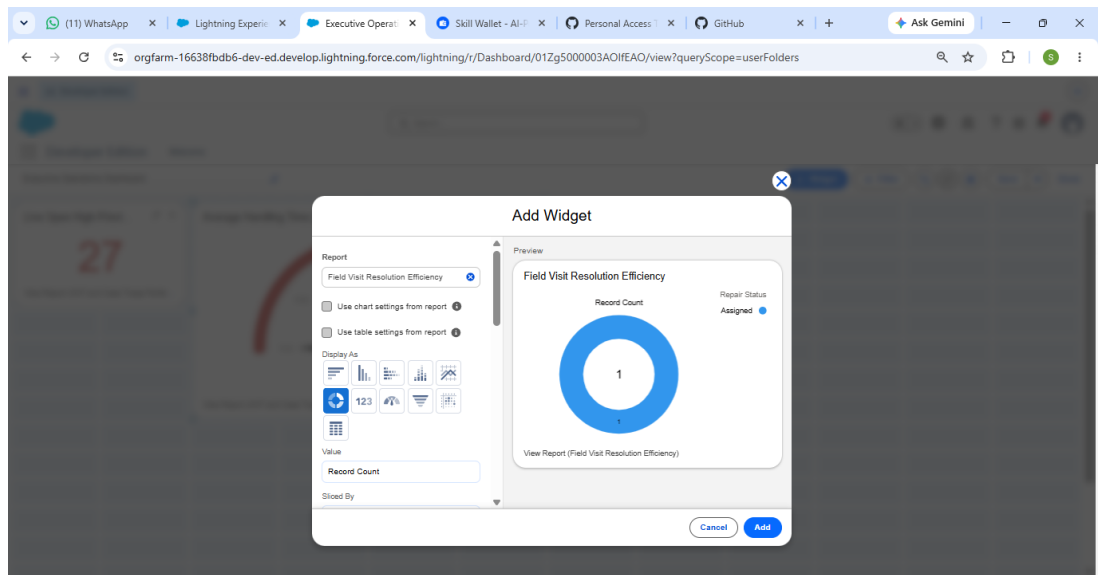
- Click **+ Component**.
- Select report: **AHT and Case Triage Performance**.
- **Configure:**
 - **Component Type:** Gauge
 - **Measure:** Average Age
 - **Breakpoint 1:** 2
 - **Breakpoint 2:** 5
 - **Title:** Average Handling Time (AHT) Gauge



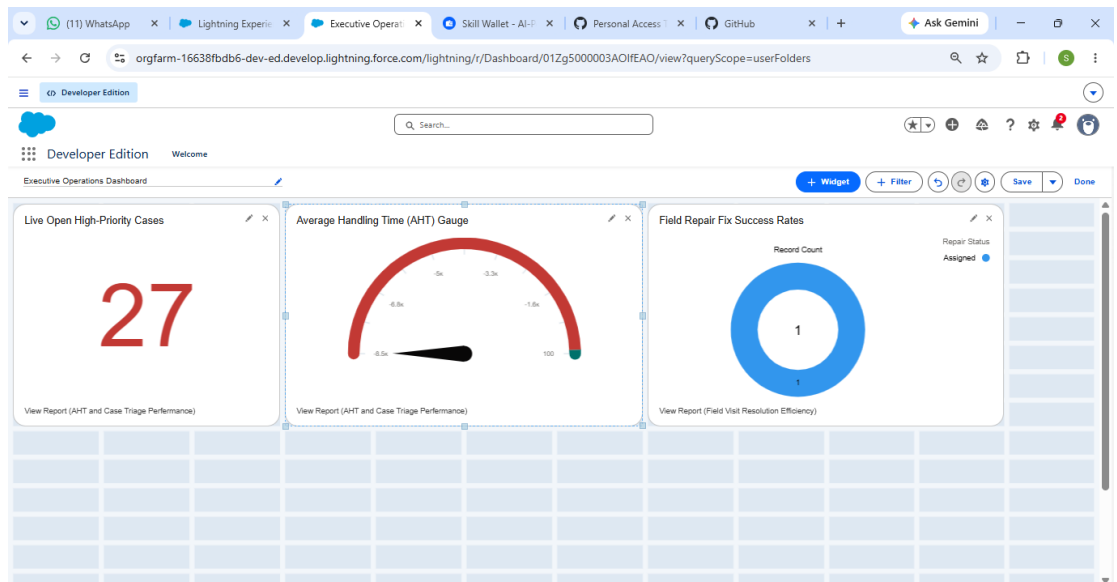
- Click **Add** and position it in the top center.

Step 8: Add Component 3 — First-Visit Fix Ratio Donut

- Click **+ Component**.
- Select: **Field Visit Resolution Efficiency**.
- Configure:
 - **Component Type:** Donut Chart
 - **Metric:** Record Count
 - **Slice By:** Repair Status
 - **Title:** Field Repair Fix Success Rates

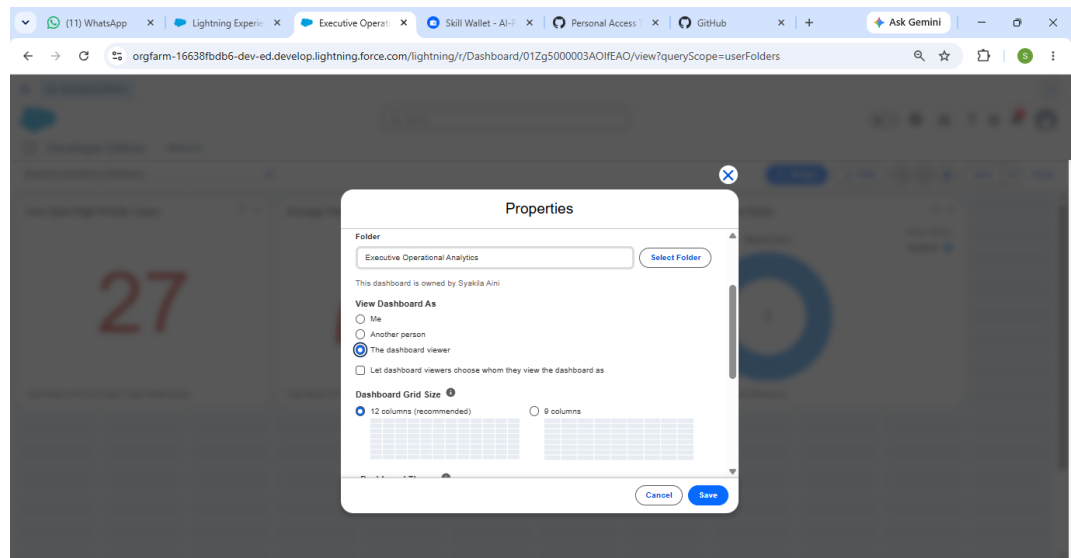


- Click **Add** and position it in the upper-right section.



Step 9: Configure Dashboard Run Context and Finalize Activation

- Open **Dashboard Settings**.
- Under **View Dashboard As**:
 - Choose **The Dashboard Viewer**, or
 - Select an **Executive Management** user.



- Click **Save**.
- Save the dashboard.
- Click **Done** to launch the active dashboard.

M8. Governance and Implementation Strategy

8.1 Governance Structure

To protect GreenGrid Utilities' Salesforce environment from data integrity issues, unauthorized system modifications, and increasing technical complexity, a formal governance model has been established. This structure supports the configuration-first approach introduced in Milestone 3 and the role-based security framework defined in Milestone 5.

8.1.1 Center of Excellence (CoE) Model

A cross-departmental Center of Excellence (CoE) is established to oversee platform performance, evaluate enhancement requests, and conduct regular security compliance reviews on a weekly basis.

- **Executive Sponsor (Executive Management Level):** Responsible for ensuring that the Salesforce platform continues to support strategic business objectives (BR1–BR3).

- **Business Change Lead (Customer Support and Field Operations Managers):** Confirms that proposed system enhancements accurately reflect operational business processes.
- **System Administrator (IT Level):** Serves as the main platform custodian, overseeing security controls, user access management, and automation optimization.

8.1.2 Environment and Release Governance

To achieve the required platform uptime target of 99.9% (NFR1), direct configuration updates in the Production environment are not permitted. Instead, a structured multi-sandbox deployment process is adopted:



8.1.3 Change Control Board (CCB) Procedures

Any updates affecting key custom objects (Utility_Asset__c, Field_Visit__c) or primary automation processes (Flow 1 and Flow 2) must pass a formal three-stage review before deployment:

1. **Risk Assessment:** Determine whether the change affects the Meter ID search performance requirement of under two seconds (FR1).
2. **Security Evaluation:** Verify that the change does not violate the Private Organization-Wide Default (OWD) model established in Milestone 5.
3. **Rollback Preparation:** A documented rollback procedure must be available to restore the previous production configuration within 15 minutes if issues occur.

8.2 User Adoption Approach

Transitioning from disconnected spreadsheet-based processes to a centralized CRM platform requires a carefully managed change strategy to reduce user resistance and eliminate data processing delays (B3).

8.2.1 Role-Specific Training Programs

Training activities should be tailored to the different user experiences designed throughout Milestones 5, 6, and 7.

- **Customer Support Program:** Interactive exercises centered on the unified Service Console. Agents practice handling customer calls, performing rapid Meter ID searches (FR1), and creating high-priority cases that trigger automated dispatch processes.
- **Field Operations Program:** Mobile-focused sessions conducted through smartphones. Technicians learn how to access the Salesforce Mobile App, respond to real-time notifications, update Repair_Status__c to "Resolved," and record Technician_Notes__c to prevent information loss.

8.2.2 Adoption Measurement and Gamification

Management evaluates implementation success through Salesforce reports that focus on three major adoption indicators:

- **Login Activity:** Ensuring all assigned support agents and field personnel access the system daily.
- **Data Quality Completion:** Reviewing closed cases to confirm Technician_Notes__c are properly completed without placeholder text or blank values.
- **Dashboard Utilization:** Measuring how frequently leadership teams access the Executive Operations Dashboard to support decision-making activities.

8.3 Deployment Roadmap

The GreenGrid Utilities Salesforce solution is implemented through a phased six-week rollout plan designed to minimize operational interruptions.

PHASED DEPLOYMENT SCHEDULE

Weeks 1–2: Stage A

Environment Lockdown & Security Foundation

Weeks 3–4: Stage B

Data Migration & Automation Validation

Weeks 5–6: Stage C

User Training, Production Launch & Hypercare Support

Stage A: Environment Lockdown and Security Foundation (Weeks 1–2)

This phase includes configuring metadata structures, enforcing Private OWD settings, defining the Role Hierarchy, and validating permission assignments within lower-level sandbox environments.

Stage B: Data Migration and Automation Validation (Weeks 3–4)

Activities include extracting legacy customer information from Excel files, indexing Meter IDs, executing duplicate data cleansing processes, and deploying the structural components of Flow 1 and Flow 2.

Stage C: User Enablement, Go-Live, and Hypercare (Weeks 5–6)

This stage covers end-user training, deployment of approved change sets to Production, execution of live smoke tests, and delivery of continuous 24/7 support during the first week following launch.

Conclusion

This project documents a complete, end-to-end Salesforce CRM transformation designed to address the core operational failures that were holding GreenGrid Utilities back. Across eight milestones, the work evolved from stakeholder discovery and process mapping all the way through data model design, security configuration, automation engineering, reporting, and long-term governance planning — each phase building directly on the one before it.

The starting point was a fragmented environment defined by three critical bottlenecks: manual customer lookups consuming up to five minutes per call, unstructured field dispatch carried out over informal messaging channels, and operational data reaching management two to four hours after jobs were completed. These were not minor inefficiencies — they were systemic gaps that directly degraded customer service quality and prevented leadership from making timely, informed decisions.

The Salesforce solution built throughout this project addresses each of those gaps in a deliberate and traceable way. Customer records are now retrieved through an indexed

Meter ID lookup in under two seconds, eliminating the spreadsheet bottleneck entirely. Field dispatch is handled automatically through a Record-Triggered Flow the moment a high-priority case is saved, removing human error from the coordination chain. And when a technician marks a job as resolved on their mobile device, the parent case closes and the executive dashboard refreshes instantly — closing the reporting gap that once took hours. Every design decision across the eight milestones was intentionally grounded in the business requirements defined at the outset. The 25% reduction target for Average Handling Time shaped how the Service Console was configured. The 99% data accuracy benchmark drove the choice of private OWD settings and the indexed External ID on the Utility Asset object. The 15% improvement target for first-visit fix rates informed the structure of the Field Visit custom object and the reporting framework built in Milestone 7. Nothing was built in isolation — each technical choice connects back to a documented business need.

The solution was also built with longevity in mind. The declarative-first architecture means the platform can be maintained, adjusted, and extended without requiring custom code development. The Center of Excellence model, Change Control Board procedures, and phased deployment roadmap defined in Milestone 8 ensure that the system will continue to perform reliably as the organization grows, without accumulating the kind of ungoverned technical debt that typically erodes CRM platforms over time.

Ultimately, this project demonstrates that a Salesforce implementation done right is not just a technology upgrade — it is a business transformation. GreenGrid Utilities moves from reactive, paper-based operations to a governed, automated, and data-driven service environment. The tools are in place, the processes are redesigned, and the foundation is built to scale.